

Homotopy Groups of Spheres

A comprehensive literature review, numerical record, and state-of-the-art report

Coverage cutoff: 8 August 2026

Classical non-equivariant homotopy groups of spheres, including unstable and stable computations, structural theorems, periodic families, geometric applications, algorithms, software, datasets, and open frontiers.

Scope of “all known results”

This report is exhaustive at the level of major structural theorem families, published complete numerical ranges, named infinite families, modern computational methods, and a source-by-source research map. It does not reproduce every isolated composition relation, differential, or generator identity from the thousands of papers in the subject; instead, it identifies the primary sources in which those relations are recorded.

Research report

Abstract

The homotopy groups $\pi_m(S^n)$ are among the most basic and least completely known invariants in topology. Their definition is elementary, but their computation has generated the Freudenthal suspension theorem, the EHP sequence, Toda brackets, the Adams and Adams-Novikov spectral sequences, generalized cohomology theories, chromatic homotopy theory, motivic and synthetic deformation methods, and some of the most sophisticated machine-assisted mathematics currently in use. This report consolidates the state of knowledge through 8 August 2026. It separates global theorems from finite-range calculations, unstable from stable information, primary from integral results, and peer-reviewed results from current preprints. The principal numerical conclusions are: the integral stable stems are completely determined through stem 83; stems 84-90 remain partially determined with explicit 2-primary ambiguities; the literature records complete stable calculations through stem 108 at $p=3$ and through stem 999 at $p=5$; the integral unstable k -stems are complete for $k \leq 20$; the 2-primary unstable k -stems are complete for $k \leq 32$; and the 2-primary 33-stem is known for $n=2, \dots, 9$ and $n \geq 28$, with $n=10, \dots, 27$ still incomplete in the most defensible reading of the 2026 source. The report also records the 2025 peer-reviewed completion of stable stems 82 and 83 and corrections to stems 70 and 71, the 2025 preprint claim resolving the last Kervaire invariant case in dimension 126, the preprint counterexamples to the telescope conjecture at every height at least two, and the rapidly expanding collection of height-two periodic families invisible to tmf . Complete numerical appendices reproduce the stable additive groups through stem 90 and Toda's integral unstable table through the 19-stem, with machine-readable CSV companions.

How to read this report

- Sections 1-4 give the foundations and the complete unstable numerical record.
- Sections 5-10 treat stable computations, named classes, chromatic structure, infinite families, and multiplicative phenomena.
- Sections 11-14 cover asymptotic size, geometry, algorithms, datasets, and the current frontier.
- Appendix A is the corrected stable additive-group table through stem 90; Appendix B is Toda's full integral unstable table through the 19-stem.
- Evidence labels distinguish peer-reviewed publications, accepted/forthcoming work, preprints, and source inconsistencies.

Contents

Executive summary	3	12. Geometric applications	17
1. Scope, conventions, and evidence standards ...	4	13. Algorithms, software, and datasets	18
2. Elementary and global structure	5	14. Current frontier and open problems	19
3. Unstable computational machinery	6	15. Chronology	20
4. The unstable numerical record	7	16. Annotated bibliography	21
5. Stable computation	9	Appendix A. Stable stems 0-90	25
6. The stable numerical record	11	Appendix B. Toda table through $k=19$	28
7. Hopf, Kervaire, and image J	12	Appendix C. Unstable 33-stem at 2	30
8. Chromatic organization	13	Appendix D. Named classes	31
9. Infinite families and periodicity	14	Appendix E. Computation ledger	32
10. Products, brackets, and detection	15	Appendix F. Companion data files	33
11. Global growth and nonvanishing	16		

Executive summary: the state of the subject in 2026

There is no closed formula for $\pi_m(S^n)$, no known algorithm competitive with specialized spectral-sequence calculations in the ranges mathematicians care about, and no finite list of families that accounts for all stable homotopy. Nevertheless, the combination of classical composition methods, chromatic structure, and modern motivic/synthetic computation gives a precise and surprisingly rich body of knowledge. The following table is the shortest accurate summary of the numerical frontier.

Object	Range	2026 status	Principal sources
Integral stable groups	$\pi_k S$ for $0 \leq k \leq 83$	Complete additive groups	IWX 2023; Burklund-Isaksen-Xu 2025 corrections/completions
Stable stems 84-90	Integral	Partial; explicit 2-primary alternatives	IWX 2023, with later corrections incorporated
Stable $p=3$	$0 \leq k \leq 108$	Classical literature reports complete computation	Tangora, Aubry, Ravenel; surveyed Wang-Xu 2023
Stable $p=5$	$0 \leq k \leq 999$	Classical literature reports complete computation	Ravenel and predecessors; surveyed Wang-Xu 2023
Integral unstable k-stems	$\pi_{n+k}(S^n)$, all $n \geq 2$, $k \leq 20$	Complete	Toda 1962; Mimura-Toda 1963
2-primary unstable k-stems	All $n \geq 2$, $k \leq 32$	Complete	Mimura; Mimura-Mori-Oda; Oda; Inoue-Mukai; Miyauchi-Mukai
2-primary unstable 33-stem	$n=2-9$ and $n \geq 28$	Known; $n=10-27$ incomplete	Oda; Miyauchi-Mukai; Yang-Wu 2026; Thomeier
Kervaire invariant one	Dimensions 2,6,14,30,62,126 only	Last case claimed resolved	Lin-Wang-Xu 2025 preprint; HHR 2016
Telescope conjecture	Heights ≥ 2 , every prime	Claimed false	Burklund-Hahn-Levy-Schlank preprint

Table 1. Numerical and structural frontier as of 8 August 2026.

The most important updates after the 2023 0-90 computation

Stable stems 82 and 83. The 2-primary components are $\pi_{82} S(2) = (Z/2)^6 + Z/8$ and $\pi_{83} S(2) = (Z/2)^3 + (Z/8)^2$.

Source: Burklund-Isaksen-Xu, Peking Mathematical Journal, 2025. Status: Published.

Corrections in stems 70 and 71. The corrected 2-primary v_1 -torsion groups are $(Z/2)^6 + Z/4$ and $(Z/2)^5 + Z/4 + Z/8$, respectively; a previously asserted hidden 2-extension in stem 70 is absent.

Source: Burklund-Isaksen-Xu, 2025. Status: Published.

Last Kervaire class. A preprint proves h_{6^2} is a permanent cycle and concludes that Kervaire-invariant-one framed manifolds exist exactly in dimensions 2, 6, 14, 30, 62, and 126.

Source: Lin-Wang-Xu, arXiv:2412.10879v2. Status: Preprint.

Telescope conjecture. A preprint constructs counterexamples at every prime and every chromatic height at least 2; height 0 and height 1 remain the positive cases.

Source: Burklund-Hahn-Levy-Schlank, arXiv:2310.17459. Status: Preprint.

New height-two families. Recent work supplies many 192-periodic 2-primary families, including classes invisible to tmf but detected $T(2)$ - and $K(2)$ -locally, plus new simple 4- and 8-torsion families and geometric consequences for exotic spheres.

Source: Bhattacharya-Bobkova-Quigley; Bobkova-Quigley; Carrick-Davies. Status: Mixed: published and preprint.

What remains unknown

Even the additive stable groups are not completely settled in stems 84-90, and essentially no contiguous integral range beyond 90 is known. Ext groups and partial Adams information extend much farther than the actual homotopy groups, but unresolved differentials and hidden extensions prevent a complete answer.

On the unstable side, the complete all-sphere 2-primary record stops at the 32-stem. The next stem remains incomplete for sphere dimensions 10 through 27. There is no analogous clean, modern, all-prime unstable table far beyond Toda's 20-stem.

Chromatic theory predicts and constructs periodic structure without classifying all elements. At every height, the gap between localized information and the actual sphere is substantial. The telescope counterexamples sharpen this distinction: finite localization and Bousfield localization no longer agree from height two onward.

The multiplicative ring π_*^*S , not merely its additive groups, is much less completely recorded. A table of groups does not determine products, Toda brackets, power operations, or the maps induced by finite complexes. Modern computations therefore maintain charts and relation databases rather than a single list of abelian groups.

1. Scope, conventions, and evidence standards

1.1 Classical scope

The primary subject is the classical based homotopy group $\pi_m(S^n)=[S^m, S^n]_*$ and its stable limit. Equivariant, motivic, and synthetic homotopy groups are included only where they furnish computations or structural explanations for the classical sphere. A genuinely exhaustive review of motivic or equivariant sphere stems would be a separate report of comparable size.

The phrase “all known results” is interpreted at the level appropriate for a literature review: every major global theorem, every published complete numerical range, all standard named infinite families, the principal multiplicative and chromatic structures, the important geometric applications, and an annotated source map for specialized relations. It is neither useful nor feasible to restate every individual Toda relation or every chart differential. Those are indexed to their primary sources and datasets.

1.2 Notation

Symbol	Meaning
$\pi_m(S^n)$	Based homotopy classes $S^m \rightarrow S^n$.
$\pi_{\{n+k\}}(S^n)$	The unstable k -stem at sphere dimension n .
π_k^*S	Stable limit $\text{colim}_n \pi_{\{n+k\}}(S^n)$; equivalently π_k of the sphere spectrum.
$G_{(p)}, G_{p^\wedge}$	p -localization and p -completion, respectively.
$\pi_k^*S(p)$	The p -primary finite subgroup in positive stems.
E, H, P	Suspension, Hopf-invariant map, and Whitehead-product map in the EHP sequence.
η, ν, σ	The stable Hopf-invariant-one classes in stems 1, 3, and 7.
α, β, γ	Greek-letter families, usually organized by chromatic height.
v_n	A periodicity operator detected at chromatic height n .
J	The stable J -homomorphism $\pi_*(O) \rightarrow \pi_*^*S$.

1.3 Evidence labels and date sensitivity

Peer-reviewed computations are treated as established, subject to explicit later corrections. Accepted or forthcoming work is identified as such. ArXiv results without a journal reference at the cutoff are called preprints even when experts broadly regard them as convincing. This matters particularly for the last Kervaire invariant problem and the all-height counterexamples to the telescope conjecture.

Numerical tables are versioned objects. Appendix A uses IWX 2023 as the base, replaces the incorrect 70- and 71-stem entries, incorporates the 2025 completion of stems 82 and 83, and retains IWX’s explicit alternatives in stems 84-90. Appendix C records the internal $n=27$ inconsistency in Yang-Wu rather than choosing the stronger caption over the contradictory prose and table.

Warning about “computed through N ”

In this literature, “computed through stem N ” may mean different things: an Ext chart exists, the E_∞ page is known, the additive p -primary groups are known, all hidden extensions are resolved, multiplicative products are known, or all primes have been assembled integrally. This report states which meaning is intended each time.

2. Elementary and global structure of $\pi_m(S^n)$

2.1 Definition, functoriality, and the first groups

For pointed spaces, $\pi_m(S^n)$ is the set of based homotopy classes of maps $S^m \rightarrow S^n$. It is a group for $m \geq 1$ and abelian for $m \geq 2$. Cellular approximation gives $\pi_m(S^n) = 0$ for $m < n$. The degree theorem gives $\pi_n(S^n) = \mathbb{Z}$, generated by the identity. The action of degree- d self-maps and composition make sphere homotopy into a network of modules and pairings rather than a collection of unrelated groups.

The first unstable deviations already display the characteristic mixture of free and torsion information. The Hopf fibration generates $\pi_3(S^2) = \mathbb{Z}$. For $n \geq 3$, $\pi_{n+1}(S^n) = \mathbb{Z}/2$, generated by a suspension of η . For every $n \geq 2$, $\pi_{n+2}(S^n) = \mathbb{Z}/2$. In the next stem the exceptional low spheres appear: $\pi_5(S^2) = \mathbb{Z}/2$, $\pi_6(S^3) = \mathbb{Z}/12$, $\pi_7(S^4) = \mathbb{Z} + \mathbb{Z}/12$, while the stable group is $\pi_3^S = \mathbb{Z}/24$.

2.2 Freudenthal stabilization

Freudenthal's suspension theorem says that $E: \pi_{n+k}(S^n) \rightarrow \pi_{n+k+1}(S^{n+1})$ is an isomorphism when $n \geq k+2$ and a surjection when $n = k+1$. Consequently, for fixed k , the k -stem becomes independent of n , and π_k^S may be identified with $\pi_{n+k}(S^n)$ in the stable range. This is the fundamental reason a single stable table controls infinitely many unstable groups.

The boundary of the stable range is mathematically rich. Immediately below it, Whitehead products and Hopf invariants contribute free summands and extensions. Thomeier's theorem recovers the first eight groups counted backward from stability from the stable stem, with congruence-dependent extra $\mathbb{Z}$ and $\mathbb{Z}/2$ summands. Modern 33-stem work still uses this 1966 result.

2.3 Serre finiteness and rational homotopy

Serre proved that $\pi_m(S^n)$ is finite except in the diagonal $m = n$ and, when n is even, in degree $m = 2n-1$. Rationally, an odd sphere has a single nonzero group: $\pi_n(S^n) \otimes \mathbb{Q} = \mathbb{Q}$. An even sphere has two: $\mathbb{Q}$ in degree n and $\mathbb{Q}$ in degree $2n-1$. The latter generator is represented rationally by the Whitehead square. Thus all positive stable stems are finite.

Because every positive stable stem is a finite abelian group, it decomposes canonically into p -primary components. Serre's first-torsion theorem implies that for odd p and $n \geq 3$, no p -torsion occurs before offset $2p-3$, and the first p -torsion is a $\mathbb{Z}/p$ class in that offset. For a fixed stable stem k , only primes p with $2p-3 \leq k$ can occur. This simple inequality is why complete integral assembly is possible after finitely many prime-local computations.

2.4 Nonvanishing and size

Finiteness is not vanishing. Ivanov-Mikhailov-Wu proved that $\pi_m(S^2)$ is nonzero for every $m \geq 2$; the Hopf fibration then gives nonvanishing of $\pi_m(S^3)$ for every $m \geq 3$. Stable groups, by contrast, do vanish in sporadic positive stems, for example stems 4, 5, 12, and 61, but there is no classification of vanishing stems.

Burklund proved that, at each prime, the torsion exponent of π_k^S grows sublinearly in k . The p -rank is bounded by $\exp(O((\log k)^3))$, hence the stable stems have subexponential size. An appendix with Senger bootstraps this to p -local unstable sphere homotopy. The best general lower bounds remain much smaller and arise from image J and periodic families; the true asymptotic growth is open.

2.5 Stable homotopy as framed bordism

The Pontryagin-Thom construction identifies π_k^S with the framed bordism group Ω_k^{fr} . A stable map is represented by the inverse image of a regular value, a k -manifold with a stable normal framing; homotopy corresponds to framed cobordism. This is not merely an interpretation: it is the bridge to exotic spheres, characteristic numbers, the Kervaire invariant, and differential topology.

Composition and smash product make π_*^S a graded-commutative ring. The additive group table records only the underlying abelian groups. Framed-manifold products, compositions among η , ν , σ , and higher Toda brackets encode additional data that remains incompletely catalogued even within the computed range.

3. Unstable computational machinery

3.1 The EHP sequence and James construction

The James construction models $\Omega\Sigma X$ by a free monoid on X up to homotopy. Applied to spheres, its Hopf construction yields the EHP fiber sequence and long exact sequence. In one standard indexing, the maps relate $\pi_{\{q+1\}}(S^{\{2n+1\}})$, $\pi_{\{q-1\}}(S^{\{n\}})$, $\pi_q(S^{\{n+1\}})$, and $\pi_q(S^{\{2n+1\}})$ through P , E , and H . Suspension moves toward stability; H measures Hopf-invariant information; P is built from Whitehead products.

Toda's tables were obtained by repeatedly threading known generators and relations through EHP. The method is inductive but not mechanical: exactness gives kernels and cokernels, while extension problems require compositions, Hopf invariants, Whitehead products, and secondary operations. The modern 33-stem extension problems are direct descendants of this methodology.

3.2 Whitehead products, Hilton-Milnor, and secondary composition

The Whitehead product $[\alpha, \beta]$ measures the failure of wedge inclusions to commute. The Whitehead square $\{\iota_n, \iota_n\}$ controls the first unstable obstruction to suspension and produces the rational class in $\pi_{\{2n-1\}}(S^{\{n\}})$ for even n . The Hilton-Milnor theorem decomposes loops on a wedge into products indexed by basic Lie words, reducing many wedge problems to sphere groups plus iterated Whitehead products.

When ordinary composition vanishes for nonunique reasons, Toda brackets record secondary composition. A bracket $\langle \alpha, \beta, \gamma \rangle$ is a coset modulo an explicit indeterminacy, and higher or matrix brackets organize longer chains. Toda brackets both construct elements and detect hidden extensions in spectral sequences. A large fraction of named stable classes - including Kervaire and Greek-letter phenomena in low filtration - are most naturally specified by such brackets.

3.3 Spectral sequences in the unstable setting

The Serre spectral sequence for path-loop fibrations and iterated loop spaces supplied Serre's early computations. The unstable Adams spectral sequence computes p -completed homotopy from unstable modules over the Steenrod algebra. The lambda algebra packages the mod-2 unstable Adams E_1 or E_2 structure and historically enabled extensive calculations. Bousfield-Kan completion, unstable modules, and Lannes theory provide additional organization.

Unlike the stable Adams spectral sequence, unstable differentials depend on the target sphere dimension and interact with EHP. Consequently, a stable chart cannot simply be desuspended. The failure of desuspension is measured by Hopf invariants, Whitehead products, and geometric boundary theorems.

3.4 Exponents and power maps

For an odd prime p , Cohen-Moore-Neisendorfer proved that p^n annihilates the p -primary torsion in all homotopy groups of $S^{\{2n+1\}}$; equivalently, a suitable p^n power map on an iterated loop space is null. This is one of the strongest uniform unstable results. Subsequent work of Gray, Anick, Neisendorfer, and others refines decompositions and exponents in special dimensions and primes.

At $p=2$, exponent behavior is subtler and lacks a comparably uniform theorem. The Moore conjecture and related exponent questions for finite complexes remain part of the broader attempt to distinguish elliptic and hyperbolic homotopy growth.

3.5 Chromatic unstable homotopy and Goodwillie calculus

The Bousfield-Kuhn functor extracts v_n -periodic unstable homotopy and returns a $T(n)$ -local spectrum. Goodwillie's Taylor tower of the identity functor approximates a space by homogeneous layers built from derivatives carrying symmetric-group actions. Arone-Mahowald showed that, for spheres, only selected layers contribute to v_n -periodic homotopy, turning an unstable problem into a structured stable calculation.

Modern work recasts the layers using spectral Lie algebras, partition complexes, and unstable chromatic categories. These methods do not yet supply an integral table, but they explain why periodic unstable families exist and how their sphere dimension controls desuspension.

4. The unstable numerical record

4.1 Complete ranges

k-stem	Localization/range	Primary source	Status
0-19	Integral, all $n \geq 1$	Toda 1962	Complete table of $\pi_{\{n+k\}}(S^n)$.
20	Integral, all n	Mimura-Toda 1963	Completes the all-prime 20-stem.
21-22	2-primary, all n	Mimura 1965	Generalized Hopf homomorphism and higher composition.
23-24	2-primary, all n	Mimura-Mori-Oda 1975	Complete 2-components.
25-31	2-primary, all $n \geq 2$	Oda 1979; later corrections	Remaining 31-stem issues resolved by Inoue-Mukai.
32	2-primary, all $n \geq 2$	Miyauchi-Mukai 2017	Complete.
33	2-primary, $n=2-9$ and $n \geq 28$	Oda; Miyauchi-Mukai; Yang-Wu; Thomeier	$n=10-27$ incomplete; $n=27$ source inconsistency.

Table 2. Complete and near-complete unstable ranges.

This ledger is more informative than saying that “homotopy groups of spheres are known to the 32-stem.” The integral record stops at 20; the extension to 32 is a 2-primary result. Odd-primary unstable calculations continue much farther in many selected dimensions and families, but they do not form a comparably clean all-sphere range.

The generator-level 2-primary tables in the Wang-Xu survey and Oda’s monograph-length paper remain indispensable. They record not only abelian groups but named generators, suspensions, Hopf invariants, P-images, and compositions needed to continue EHP calculations.

4.2 Toda’s table through the 19-stem

Appendix B reproduces the complete integral table from Toda’s Composition Methods, organized by offset k and sphere dimension n . It is the largest compact all-prime unstable table that remains standard. The entries exhibit stabilization diagonally, free $\mathbb{Z}$ summands in the Whitehead-square degree for even spheres, and the rapid proliferation of 2-primary summands.

A table entry such as $8+2+3$ means $\mathbb{Z}/8 + \mathbb{Z}/2 + \mathbb{Z}/3$, “infy” means $\mathbb{Z}$, and 1 means the trivial group. The CSV companion preserves this compact notation so it can be compared against online databases or transformed programmatically.

4.3 Selected exact low-stem formulas

Case	Group/theorem	Range
k=0	$\pi_n(S^n) = \mathbb{Z}$	All $n \geq 1$
k=1	$\pi_2(S^1) = 0$; $\pi_3(S^2) = \mathbb{Z}$; $\pi_{\{n+1\}}(S^n) = \mathbb{Z}/2$	$n \geq 3$ for stable formula
k=2	$\pi_{\{n+2\}}(S^n) = \mathbb{Z}/2$	All $n \geq 2$
k=3	$\pi_5(S^2) = \mathbb{Z}/2$; $\pi_6(S^3) = \mathbb{Z}/12$; $\pi_7(S^4) = \mathbb{Z} + \mathbb{Z}/12$; stable group $\mathbb{Z}/24$	Exceptional desuspensions
rational	$\pi_m(S^n) \otimes \mathbb{Q} = \mathbb{Q}$ for $m=n$, and also $m=2n-1$ when n even; otherwise 0	Serre
first odd-p torsion	No p -torsion for $k < 2p-3$; a $\mathbb{Z}/p$ occurs at $k=2p-3$	$n \geq 3$, p odd

4.4 The 2-primary 33-stem

Yang-Wu resolve the three Oda extension problems by constructing an element κ'_6 of order four and tracking its suspensions. Their final groups are $\pi_{39}(S^6)_2 = \mathbb{Z}/4 + (\mathbb{Z}/2)^6$, $\pi_{40}(S^7)_2 = \mathbb{Z}/4 + (\mathbb{Z}/2)^4$, and $\pi_{41}(S^8)_2 = \mathbb{Z}/8 + \mathbb{Z}/4 + (\mathbb{Z}/2)^6$.

Their Table 1 combines these with earlier calculations and Thomeier’s backward theorem. However, the surrounding prose says the cases $10 \leq n \leq 27$ are still being worked on, while the caption and corollary state $n \geq 27$; the table itself begins at $n=28$. Until an erratum or later paper resolves this, the conservative complete range is $n \geq 28$, with $n=27$ unresolved.

n	$\pi_{\{n+33\}}(S^n)$ at 2	Basis/status
2	$\mathbb{Z}/4 + (\mathbb{Z}/2)^2$	Oda; summarized by Yang-Wu
3	$(\mathbb{Z}/2)^3$	Oda; summarized by Yang-Wu

n	$\pi_{\{n+33\}}(S^n)$ at 2	Basis/status
4	$(\mathbb{Z}/8)^2 + (\mathbb{Z}/2)^6$	Oda; summarized by Yang-Wu
5	$(\mathbb{Z}/2)^4$	Oda; summarized by Yang-Wu
6	$\mathbb{Z}/4 + (\mathbb{Z}/2)^6$	Yang-Wu 2026
7	$\mathbb{Z}/4 + (\mathbb{Z}/2)^4$	Yang-Wu 2026
8	$\mathbb{Z}/8 + \mathbb{Z}/4 + (\mathbb{Z}/2)^6$	Yang-Wu 2026
9	$(\mathbb{Z}/2)^6$	Miyauchi-Mukai 2017
27	unresolved / source discrepancy	Yang-Wu text says project covers 10-27; caption says $n \geq 27$
28	$(\mathbb{Z}/2)^5$	Thomeier backward theorem + stable 33-stem
29	$(\mathbb{Z}/2)^5$	same
30	$(\mathbb{Z}/2)^5$	same
31	$(\mathbb{Z}/2)^6$	same
32	$(\mathbb{Z}/2)^7$	same
33	$(\mathbb{Z}/2)^6$	same
34	$(\mathbb{Z}/2)^5 + \mathbb{Z}_2(2)$	same
35	$(\mathbb{Z}/2)^5$	stable and all $n \geq 35$

Table 3. Known 2-primary groups in the unstable 33-stem; $n=27$ is deliberately flagged.

4.5 Odd-primary unstable information

Odd-primary unstable homotopy is structurally cleaner in several respects: the first torsion is later, odd spheres admit useful p -local splittings, and power maps have strong exponent bounds. Toda, Mimura, Oka, and many later authors computed extensive α - and β -family desuspensions, EHP differentials, and homotopy groups of Lie groups. These results are distributed across specialized papers rather than a single modern complete table.

At an odd prime, the α -family begins in stable stem $2p-3$ and has period $2p-2$ in its height-one form. Unstable representatives exist only above dimension-dependent bounds. The β -family is height two and interacts with double suspension, Moore spaces, and finite H -spaces. Ravenel's book and the Wang-Xu survey are the best entry points to the classical computations; modern work tends to formulate odd-primary information chromatically rather than as all-sphere tables.

4.6 Why the unstable frontier advances slowly

Each new unstable stem requires more than knowing the stable endpoint. One must determine how far each stable generator desuspends, compute EHP kernels and images, resolve group extensions, and prove relations among named unstable representatives. A single unresolved order-four versus two-copies-of-order-two extension changes the group even when every associated graded piece is known.

The 2026 paper on relations in the 24th homotopy groups illustrates the dependency chain: relations established in one offset may be prerequisites for a much higher stem because compositions with σ , κ -bar, and generalized P -images propagate. Thus “old” low-stem identities can remain the bottleneck for new calculations decades later.

5. Stable computation: from Adams to synthetic spectra

5.1 The stable category and the sphere spectrum

Stabilization formally inverts suspension and replaces spaces by spectra. The sphere spectrum S is the tensor unit, and $\pi_k \wedge S = [\Sigma^k S, S]$. Negative groups vanish because the classical sphere spectrum is connective; $\pi_0 \wedge S = \mathbb{Z}$; positive groups are finite. Stable maps admit cofiber sequences, exact triangles, Spanier-Whitehead duality, generalized homology and cohomology theories, and spectral sequences unavailable in the same form unstably.

The computational problem is nevertheless difficult because S is the initial E_∞ ring spectrum and therefore contains every chromatic layer. A generalized cohomology theory sees only part of S . Stable homotopy calculations proceed by covering or resolving S with spectra whose cooperations are algebraically tractable, then reconstructing the missing extensions and multiplicative structure.

5.2 The Adams spectral sequence

At a prime p , the mod- p Adams spectral sequence has $E_2^{s,t} = \text{Ext}_A^{s,t}(F_p, F_p)$ and converges to the p -completed stable stem $\pi_{t-s} \wedge S$. The filtration s measures the complexity of a class with respect to ordinary mod- p cohomology. The Steenrod algebra A is explicitly known but enormous; computing Ext requires resolutions, the lambda algebra, the May spectral sequence, secondary operations, and substantial computer algebra.

An E_∞ chart is only an associated graded object. To recover $\pi_* \wedge S$ one must determine hidden p -extensions and hidden multiplications by η , ν , and other classes. Differentials can be forced by naturality through finite complexes, Kahn-Priddy transfer, geometric boundary theorems, comparison with generalized cohomology theories, or Toda-bracket convergence. The 61-stem and 82-83 stem proofs are paradigmatic examples.

5.3 May and secondary Steenrod machinery

The May spectral sequence filters the Steenrod algebra and computes its Ext from a more accessible graded object. It identifies families such as h_i , $b_{i,j}$, and their products, but then introduces its own differentials and hidden multiplicative extensions. Bruner's computer resolution and the Bruner-Rognes Ext database make the low-dimensional E_2 page reproducible.

Secondary Steenrod algebras and Baues-Jibladze or Nassau models seek to compute Adams d_2 and higher structure algebraically. Chua's modern work on Adams differentials via secondary operations and the Lin-Wang-Xu machine framework continue this program. These approaches are especially valuable because spectral-sequence charts contain many differential possibilities that cannot be excluded by degree alone.

5.4 The Adams-Novikov spectral sequence and chromatic filtration

Complex cobordism MU and its p -typical summand BP replace the Steenrod algebra by the Hopf algebroid of formal group laws. The Adams-Novikov spectral sequence has $E_2 = \text{Ext}_{BP_*BP}^{s,t}(BP_*, BP_*)$ and converges to the p -local stable stems. Its filtration is often closer to chromatic complexity: image- J classes lie at height one, β -families at height two, and higher Greek-letter families at increasing heights.

The chromatic spectral sequence resolves the moduli stack of formal groups by height strata. Miller-Ravenel-Wilson used it to construct periodic families and Ravenel systematized the subject. At small primes the algebra remains formidable, but the Adams-Novikov viewpoint explains periodicity and supports the long classical $p=3$ and $p=5$ computations.

5.5 bo-, tmf-, and finite-resolution methods

At height one, real K -theory and the connective image-of- J spectrum j organize v_1 -periodic information and isolate v_1 -torsion. Mahowald's bo-resolutions and Davis-Mahowald techniques produced many 2-primary calculations and vanishing curves. At height two, topological modular forms tmf and resolutions by spectra with level structure detect modular and periodic families.

A generalized cohomology theory can both detect and fail to detect. A class with nonzero tmf -Hurewicz image is certified, but a class invisible to tmf may still survive in the sphere. Recent 192-periodic families are specifically important because they demonstrate systematic height-two homotopy beyond ordinary tmf detection.

5.6 Motivic deformation and the cofiber of tau

Over $\mathbb{C}$, motivic stable homotopy introduces a weight grading and an element τ . Inverting τ recovers classical stable homotopy, while the cofiber $C\tau$ has an algebraic homotopy category closely related to BP_*BP comodules. This turns classical Adams-Novikov information into a geometric deformation parameter and exposes hidden extensions that are difficult to see classically.

Isaksen used the motivic May and Adams spectral sequences to compute through stem 59. Wang-Xu and Isaksen-Wang-Xu combined motivic charts, cofiber-of-tau comparisons, finite-complex arguments, and machine Ext calculations to reach 90. The method is not merely “compute motivically and invert tau”: one must prove motivic differentials, compare filtrations, and solve tau-, 2-, η -, and ν -extensions.

5.7 Synthetic spectra and the 2025 corrections

Synthetic spectra provide another deformation of stable homotopy, with a parameter λ that records Adams filtration history. A classical d_r differential corresponds to λ^{r-1} -torsion in the synthetic category. This extra structure can prove differentials and extensions by constructing finite synthetic complexes and maps whose classical shadows force the desired behavior.

Burklund-Isaksen-Xu use F_2 -synthetic methods to prove that $h_6 g + h_2 e_2$ is a permanent cycle, completing the 82- and 83-stems. They also incorporate machine-discovered corrections to a d_5 differential, changing the 70- and 71-stem groups. This episode demonstrates why even peer-reviewed tables require versioned correction ledgers.

5.8 Historical computation frontier

Period	Principal authors	Milestone
1951-1953	Serre	Stable stems below 9 using loop-space homology and spectral sequences.
1962	Toda	Integral stable and unstable information through the 19-stem.
1964-1965	May and early Adams school	Ext computations and stable range through roughly 28.
1967-1984	Mahowald, Tangora, Barratt, Bruner, others	2-primary classical computations into the 40s; extensive odd-primary work.
1990-1995	Kochman; Kochman-Mahowald	Computer-assisted consolidation through approximately stem 47.
2015-2019	Isaksen and collaborators	Motivic 2-primary computation through 59; corrections in 51-52.
2017	Wang-Xu	61-stem vanishing.
2020-2023	Isaksen-Wang-Xu	Mostly complete 2-primary computation through 90.
2025	Burklund-Isaksen-Xu	Complete 82-83; correct 70-71.

Table 4. A compressed history of the stable numerical frontier.

6. The stable numerical record

6.1 Exact integral groups through stem 83

Appendix A gives the additive decomposition of every integral stable stem through 90. Through stem 83, the groups are exact after incorporating the 2025 corrections. “Exact” here means the finite abelian group is determined. It does not mean that every generator has a canonical name or that every product and Toda bracket is known.

IWX organize each group into 2-primary v1-torsion, odd-primary v1-torsion, and a v1-periodic subgroup. This decomposition is computationally and chromatically informative, though not always a canonical decomposition of rings. The table in this report preserves the columns and also prints the total additive group.

6.2 Stems 84-90

Stem	2-primary v1-torsion	Odd v1-torsion	v1-periodic	Status
84	$(\mathbb{Z}/2)^6$ OR $(\mathbb{Z}/2)^5$	$(\mathbb{Z}/3)^2$	0	partial: one unresolved $\mathbb{Z}/2$
85	$(\mathbb{Z}/2)^6 + (\mathbb{Z}/4)^2$ OR $(\mathbb{Z}/2)^5 + (\mathbb{Z}/4)^2$ OR $(\mathbb{Z}/2)^4 + (\mathbb{Z}/4)^3$ OR $(\mathbb{Z}/2)^7 + \mathbb{Z}/4$	$(\mathbb{Z}/3)^2$	0	partial: extension/differential ambiguity
86	$(\mathbb{Z}/2)^4 + (\mathbb{Z}/8)^2$ OR $(\mathbb{Z}/2)^2 + \mathbb{Z}/4 + (\mathbb{Z}/8)^2$	$\mathbb{Z}/3 + \mathbb{Z}/5$	0	partial: 2-primary ambiguity
87	$(\mathbb{Z}/2)^5 + \mathbb{Z}/4$	0	$\mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/23$	exact
88	$(\mathbb{Z}/2)^4 + \mathbb{Z}/4$	0	$\mathbb{Z}/2$	exact
89	$(\mathbb{Z}/2)^3$	0	$(\mathbb{Z}/2)^2$	exact
90	$(\mathbb{Z}/2)^3 + \mathbb{Z}/8$ OR $(\mathbb{Z}/2)^2 + \mathbb{Z}/8$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	partial: one unresolved $\mathbb{Z}/2$

Table 5. Explicit unresolved alternatives in the top of the 0-90 range.

The ambiguity in stem 84 is one possible $\mathbb{Z}/2$. Stem 85 has several possible extension patterns, and stem 86 differs by a $\mathbb{Z}/2$ versus $\mathbb{Z}/4$ contribution. Stem 90 again has one possible $\mathbb{Z}/2$. By contrast, the additive groups in stems 87, 88, and 89 are exact. Thus the uncertainty inside the 84-90 window is localized rather than present in every row.

There is information beyond stem 90 - Ext classes, individual differentials, periodic elements, Kervaire-related classes in 126, and localized families - but not a complete contiguous table of integral stable groups. The existence of a class in stem 126 does not imply knowledge of π_{126}^S as an abelian group.

6.3 Odd-prime numerical ranges

The classical literature, as summarized by Wang-Xu, records the $p=3$ stable groups through stem 108 and the $p=5$ groups through stem 999. These long ranges are possible because odd-primary Steenrod and Adams-Novikov structures are sparser, and because Ravenel’s chromatic calculations become especially effective at $p \geq 5$. They should not be confused with integral ranges: the 2-primary component dominates the difficulty in low stems.

For primes $p \geq 7$, general vanishing lines and sparse Greek-letter calculations cover large regions, and any fixed low stem contains no p -torsion once $2p-3$ exceeds the stem. There is not a single standard “complete through N ” claim for every larger prime analogous to $p=3$ and $p=5$. The practical source is Ravenel’s green book together with later correction and family papers.

6.4 Multiplicative information is a separate frontier

The ring π_*^S is graded-commutative, and composition products can be nonzero even when additive generators are ambiguous. Named relations such as $\eta\nu=0$ or $\nu\sigma$ -type products, orders of κ and $\bar{\kappa}$ -families, and products among periodic classes are distributed across Toda, Adams, Mahowald, Ravenel, tmf calculations, and current family papers.

Spectral-sequence charts retain products and filtrations that an abelian-group table discards. For research use, Appendix A should be paired with the IWX Adams charts and their Zenodo data, the Bruner-Rognes Ext database, and machine-proof datasets. A complete multiplication table through 90 has not been published as a single audited object.

7. Distinguished classes: Hopf invariant, Kervaire invariant, and image J

7.1 The Hopf-invariant-one theorem

The classical Hopf invariant assigns an integer to maps $S^{2n-1} \rightarrow S^n$. Hopf fibrations give invariant one for $n=1,2,4,8$. Adams proved these are the only possibilities. Stably, the nontrivial positive-dimensional Hopf-invariant-one classes are η in $\pi_1 S$, ν in $\pi_3 S$, and σ in $\pi_7 S$.

Their orders in the integral stable groups are 2, 24, and 240, respectively. Their 2-primary orders are 2, 8, and 16. These classes generate much of the low-dimensional algebra, but Nishida nilpotence implies each positive-degree class, including η , ν , and σ , becomes zero after sufficiently many products.

7.2 Kervaire-invariant-one classes

Class	Stem	Description	Status/source
θ_1	2	η^2	Classical
θ_2	6	ν^2	Classical
θ_3	14	σ^2	Classical
θ_4	30	Higher class	Barratt-Mahowald-Tangora era
θ_5	62	Higher class	Barratt-Jones-Mahowald / Lin / Xu
θ_6	126	Detected by h_6^2	Lin-Wang-Xu 2025 preprint
$\theta_j, j \geq 7$	$2^{j+1}-2$	Do not exist	Hill-Hopkins-Ravenel 2016

Browder proved that Kervaire-invariant-one elements can occur only in stems $2^{j+1}-2$ and correspond in the Adams spectral sequence to h_j^2 . Hill-Hopkins-Ravenel ruled out all $j \geq 7$, leaving $j=6$, stem 126. Lin-Wang-Xu's current preprint proves h_6^2 survives. Because the last step is still a preprint at the report cutoff, the mathematically accurate formulation is "claimed resolved with a detailed proof and machine-supported companion," not "long-published theorem."

The Kervaire problem exemplifies the distinction between detecting one element and computing a stem. The claimed θ_6 settles a yes/no geometric invariant without determining the entire 126-stem.

7.3 The stable J-homomorphism

Bott periodicity computes $\pi_*(O)$, and the stable J-homomorphism sends a stable vector bundle clutching function to a framed sphere. Adams computed its image. In positive stems congruent to 0 or 1 modulo 8, the classical image of J has order 2. In stem $4m-1$, it is cyclic of order equal to the denominator of $B_{2m}/(4m)$, where B_{2m} is a Bernoulli number. It is zero in the other positive congruence classes allowed by Bott periodicity.

Examples are order 24 in stem 3 and order 240 in stem 7. At an odd prime p , the p -primary image in degree $2(p-1)r-1$ has order $p^{v_p(r)+1}$. The Adams e-invariant detects these classes. Image J supplies an infinite height-one family and a logarithmic lower bound on torsion exponents.

7.4 Full v_1 -periodic homotopy versus image J

At odd primes, positive v_1 -periodic sphere homotopy is essentially governed by image J. At $p=2$, the full $K(1)$ -local or v_1 -periodic contribution also includes classes such as η^2 not literally in the classical image of J. Modern stable tables therefore separate a v_1 -periodic column from v_1 -torsion rather than labeling the entire column "image J."

The $K(1)$ -local sphere can be modeled as the fiber of ψ^q-1 on p -complete K-theory for a topological generator q of $\mathbb{Z}_p^\times$, with modifications at $p=2$. This converts height-one stable stems into p -adic valuations of q^r-1 and explains their rigid periodic arithmetic.

8. Chromatic organization of the stable stems

8.1 Formal groups and height

Complex-oriented cohomology theories assign formal group laws to tensor products of line bundles. Over a field of characteristic p , one-dimensional formal groups have heights $1, 2, \dots$ or infinity. Morava K -theory $K(n)$ isolates height n ; Morava E -theory E_n is a universal deformation theory near a height- n formal group. The chromatic philosophy decomposes p -local stable homotopy into layers indexed by height.

Height zero is rational homotopy; height one is K -theory and image J ; height two is elliptic or modular and is partially seen by tmf ; higher heights require increasingly complicated automorphism groups and descent. The sphere contains all heights simultaneously, so chromatic localization provides approximations rather than a finite decomposition.

8.2 Nilpotence theorem

Nishida proved directly that every positive-degree element of π_*^S is nilpotent. Devinatz-Hopkins-Smith generalized this: a map of finite spectra is smash-nilpotent precisely when every Morava K -theory annihilates it. This transforms a difficult homotopy-ring property into a family of field-valued homology tests.

Nilpotence does not mean scarcity. A class can generate arbitrarily long periodic families after acting on a finite complex, even though its product in the sphere ring eventually vanishes. The distinction between nilpotent sphere elements and periodic self-maps of finite spectra is central to chromatic theory.

8.3 Thick subcategories and periodicity

Hopkins-Smith classified thick subcategories of finite p -local spectra: they form the chain $C_0 \supset C_1 \supset C_2 \supset \dots$, where C_n consists of spectra of type at least n , meaning $K(i)_* X = 0$ for $i < n$. Their periodicity theorem gives every type- n finite spectrum a v_n self-map, unique up to iterates in an asymptotic sense.

The theorem guarantees periodic families in the homotopy of finite spectra. Producing sphere classes from them requires additional constructions - boundary maps, root invariants, Greek-letter constructions, transfers, or detection by localized units. This is why periodicity is structural but does not by itself enumerate π_*^S .

8.4 Chromatic convergence and fracture

For a finite p -local spectrum X , chromatic convergence reconstructs X from its $E(n)$ -localizations $L_n X$ via a homotopy limit. Fracture squares assemble adjacent heights. The monochromatic layer $M_n X$ is the fiber of $L_n X \rightarrow L_{n-1} X$ and isolates height n . These towers organize both theory and computations.

Convergence is not an explicit algorithm: one must compute local categories, descent spectral sequences, and gluing maps. At height two and small primes, tmf and Morava stabilizer group cohomology provide partial access. At height three and above, even localized sphere groups are known only in restricted ranges and families.

8.5 The telescope conjecture

For a type- n finite spectrum with a v_n self-map, one may invert the self-map to form a telescope $T(n)$. Ravenel's telescope conjecture asserted that localization with respect to $T(n)$ agrees with $K(n)$ -localization, or equivalently that finite and Bousfield chromatic localizations coincide at height n . It is true at heights 0 and 1.

Burklund-Hahn-Levy-Schlank construct K -theoretic counterexamples at every prime and every height at least 2. As of the cutoff, this is an arXiv preprint rather than a journal-published theorem, but it has already changed the conceptual landscape. Periodic families detected $T(n)$ -locally need not exhaust $K(n)$ -local behavior, and unstable Bousfield-Kuhn constructions must keep telescopic and chromatic variants distinct.

8.6 Chromatic splitting and unresolved reconstruction

The chromatic splitting conjecture and its refinements predict how L_{n-1} information sits inside the $K(n)$ -local sphere. Counterexamples and corrections at higher heights show that naive splitting patterns fail. The modern problem is not whether chromatic layers exist, but how complicated their extensions and descent data are.

The failure of the telescope conjecture also affects asymptotic expectations. Burklund's growth paper identifies a precise strong form of telescope failure that would make the Adams E_2 upper bound essentially sharp. Thus the numerical size of the stable stems is linked to the geometry of chromatic localization.

9. Infinite families and periodic phenomena

9.1 Height-one families

The image-of-J family is the prototypical infinite family. At odd p , Toda's α -family lies in stems congruent to -1 modulo $2(p-1)$, with orders controlled by p -adic valuations. At $p=2$, Bott periodicity creates an eightfold pattern containing η -, μ -, and J -related classes. The connective image-of-J spectrum j packages these systematically.

Recent synthetic work proves strong detection statements for the unit $S \rightarrow j$, clarifying that the v_1 -periodic portion is visible in a manageable target. The remaining stable stems are v_1 -torsion and contain all height-two and higher phenomena.

9.2 Greek-letter families

The Greek-letter construction starts from regular sequences in BP_* and boundary maps in generalized Moore spectra. The α -family is height one; β is height two; γ and higher letters reflect higher chromatic layers. Miller-Ravenel-Wilson established broad existence and nontriviality results at suitable primes, while low-prime corrections require detailed differentials and divided Greek-letter variants.

Greek-letter notation is not globally uniform. A symbol may denote an Adams-Novikov class, a chosen homotopy representative, or a divided family member with indeterminacy. Reliable use requires specifying the prime, indexing convention, and detector.

9.3 Mahowald families and root invariants

Mahowald's root invariant uses projective spectra and filtered transfers to raise chromatic height: an input class often has a root invariant represented by a higher-height family. This mechanism predicts β -family representatives from α -classes and height-two families from low Hopf classes. It is powerful but technically delicate because root invariants are cosets and filtered variants can differ from actual homotopy classes.

Mahowald uncertainty principles, as emphasized by Isaksen-Wang-Xu, say informally that high-filtration classes can force differentials or extensions far away in the chart. Computations therefore propagate nonlocally; a missing relation in one stem may obstruct a distant result.

9.4 Height two at the prime 2: 192-periodicity

At $p=2$, the natural periodicity visible through tmf and related spectra is 192 in topological degree. Classical families include powers or translates of κ -bar and modular-discriminant phenomena. Not every 192-periodic sphere class is seen by tmf , and the kernel of the tmf Hurewicz map has become a major source of new families.

Bhattacharya-Bobkova-Quigley identify seven new 192-periodic families that are zero in tmf but nonzero after both $T(2)$ - and $K(2)$ -localization. They also determine new 2-torsion and divisibility properties of known families. Bobkova-Quigley add five simple η -torsion families. Carrick-Davies find families in nineteen residue classes modulo 192, including seven simple 4-torsion and four simple 8-torsion classes, detected by fixed points of tmf under an Atkin-Lehner involution.

9.5 Height two at the prime 3

The 3-primary β -family and its products are central height-two phenomena. Oka-Toda and Ravenel established the classical foundations. Recent Carrick-Davies work proves many products in v_2 -periodic families are nonzero and develops a connective height-two image-of-J analogue. Davies's analysis of β_{-1} shows β_{-1}^5 is nonzero while β_{-1}^6 vanishes, illustrating that nilpotence and periodic translation coexist.

The natural modular periodicity at $p=3$ is often 144. As at $p=2$, localized periodic families do not automatically determine the integral stable groups in each residue class; hidden extensions and higher-height interference remain.

9.6 Higher-height families

At primes large relative to height, Greek-letter and Smith-Toda-complex methods construct v_n -periodic classes for arbitrarily large n . At small primes, the existence of finite type- n complexes and sufficiently structured self-maps is more subtle, and some classical Smith-Toda complexes fail to exist. Nonetheless, nilpotence and periodicity guarantee height- n phenomena in the finite stable category.

The sphere-level problem is to detect those phenomena in the unit. Red-shift in algebraic K -theory, topological automorphic forms, and higher real K -theories offer possible detectors. No complete list of higher-height sphere families is known, and "chromatic redshift produces a class" is generally a research program rather than an automatic theorem.

10. Products, Toda brackets, Hurewicz images, and detection

10.1 Additive groups do not determine the stable homotopy ring

Two spectral-sequence survivors in adjacent stems may multiply to zero, to a named class, or to a class hidden in higher filtration. Graded commutativity imposes 2-torsion constraints on odd-degree squares but leaves many products undecided. Toda's composition tables, tmf Hurewicz calculations, and recent periodic-product papers are therefore independent layers of knowledge beyond Appendix A.

Relations also depend on choices of representatives. Toda brackets are sets with indeterminacy, and a statement that a bracket “contains x ” is weaker than a strict equality. Modern computational papers increasingly state explicit choices of lifts and finite-cell complexes so that products become machine-checkable.

10.2 Stable Hurewicz homomorphisms

The ordinary stable Hurewicz map from $\pi_k \wedge S$ to $H_k(S; \mathbb{Z})$ is zero for $k > 0$ because the sphere spectrum has homology only in degree zero. Useful Hurewicz maps therefore target generalized homology theories such as ko , j , tmf , $BP\langle n \rangle$, or Morava E-theory. A nonzero image detects a sphere class and often identifies its chromatic height.

The ko/j image captures height one. The 2-primary tmf Hurewicz image, computed by Behrens-Mahowald-Quigley, detects a rich but incomplete set of height-two classes. New families in the kernel show that no single connective theory provides a complete detector.

10.3 The Curtis conjecture and spherical classes

The stable adjoint maps $\pi_* \wedge S$ into $H_*(QS^0; F_2)$, where Dyer-Lashof operations create an enormous algebra. The Curtis conjecture predicts that, on positive-degree indecomposables, only Hopf-invariant-one and Kervaire-invariant-one classes are spherical. Many partial results bound Adams filtration, primitive length, or families, but the full conjecture remains open.

The conjecture should not be confused with the ordinary stable Hurewicz map to $H_*(S)$. It concerns the unstable homology of the infinite loop space QS^0 and is one of the sharpest proposed characterizations of geometrically visible stable classes.

10.4 Kahn-Priddy transfer

The transfer $\Sigma^\infty RP^\infty_+ \rightarrow S$ induces a surjection onto the positive-dimensional 2-primary stable stems. Algebraically, this relates Ext for the sphere to Ext for projective spectra. Geometrically, it converts sphere questions into cell-by-cell questions on RP^∞ . Wang-Xu's 61-stem proof uses an algebraic and geometric Kahn-Priddy mechanism to force the differential $d_3(D_3) = B_3$.

Balderrama's 2026 norm-based proof reveals that Kahn-Priddy is a manifestation of a more general multiplicative norm phenomenon and extends it to localized, motivic, and synthetic settings.

10.5 Infinite Adams-line families

The Adams 1-line contains h_i candidates; Adams's theorem kills all but the Hopf classes by differentials. The 2-line contains h_i^2 and the Kervaire problem. The 3-line h_j^3 supports an infinite family of d_4 differentials determined by Burklund-Xu. The 4-line includes the e -family; Li-Li's 2026 result proves the predicted differential pattern known as the New Doomsday Conjecture for that family.

These results are “all j ” theorems inside a fixed filtration, complementary to finite-stem tables. They show how a small algebraic pattern can govern infinitely many potential sphere elements even when the full stems are unknown.

11. Global growth, torsion, and nonvanishing

11.1 Torsion exponent

For a finite p -group G , the p -exponent is the least p^e annihilating G . Adams vanishing lines gave linear bounds on the exponent of $\pi_k^S(p)$; bo-resolution and BP<1> methods improved the coefficient. Burklund's theorem removes the linear term: the logarithmic p -exponent grows $o(k)$. This is currently the strongest general qualitative bound.

Image J gives elements of order $p^{v_p(r)+1}$, so the exponent is at least logarithmic along suitable stems. No construction is known that systematically produces orders near the sublinear upper bound. Hidden p -extensions in the Adams-Novikov spectral sequence are the only possible source of substantially larger orders.

11.2 Rank and total size

The May E_1 page gives $\log \text{rank}_p \pi_k^S = O((\log k)^3)$, hence rank and group size are subexponential. Burklund proves that the cumulative Adams Ext and Adams-Novikov E_2 ranks actually have $\exp(\Theta((\log k)^3))$ growth. Differentials might drastically reduce this, but a precise strong telescope-failure scenario would imply the upper bound is essentially sharp.

For unstable p -local groups of a fixed sphere, stable estimates combined with EHP and loop-space arguments yield $\exp(O((\log m)^3))$ size bounds in degree m . This is the first general subexponential upper bound, but it is far from the sporadic numerical data.

11.3 Vanishing and gaps

Stable vanishing stems occur, but no theorem classifies them. Periodic families imply infinitely many nonzero stems in specified congruence classes. The image of J alone gives nonzero groups in infinitely many stems 0,1,3,7 modulo 8 patterns, and higher chromatic families fill many others.

The stable 61-stem is zero, a result requiring a long differential proof rather than a general vanishing criterion. In contrast, unstable $\pi_m(S^2)$ and $\pi_m(S^3)$ never vanish in their positive ranges. This sharp difference cautions against extrapolating stable gaps to fixed spheres.

12. Geometric applications

12.1 Homotopy spheres and smooth structures

For $n \geq 5$, oriented homotopy n -spheres form a finite abelian group Θ_n under connected sum. Kervaire-Milnor relate Θ_n to $\text{bP}_{\{n+1\}}$, the subgroup bounding parallelizable manifolds, and to $\text{coker } J$ in $\pi_n S$, with a Kervaire-invariant correction in the relevant congruence class. Thus stable stem computations translate into counts of differentiable structures on topological spheres.

The exact sequence has congruence-dependent form, so it is unsafe to quote a single short exact sequence in every dimension. Practically, one computes $\text{image } J$, $\text{coker } J$, the Kervaire map, and bP orders separately. IWX's fourth table column performs this assembly through dimension 90 except dimension 4.

12.2 Unique smooth structures

Kervaire-Milnor established uniqueness in dimensions 5, 6, and 12. Isaksen proved uniqueness in dimension 56, and Wang-Xu in dimension 61. IWX formulate the conjecture that, among dimensions greater than 4, these are the only spheres with unique smooth structures. Dimension 4 remains outside high-dimensional surgery theory and the smooth four-dimensional Poincaré problem is open.

For odd dimensions, Wang-Xu's 61-stem calculation completes the unique-sphere classification: the only odd-dimensional spheres with unique smooth structure are S^1 , S^3 , S^5 , and S^{61} . The statement includes low-dimensional differential topology, not only stable stems.

12.3 Exotic-sphere detection by chromatic classes

Behrens-Hill-Hopkins-Mahowald use $\text{coker } J$ and generalized cohomology to detect exotic spheres in low dimensions. Height-two periodic families yield infinite geometric consequences: Carrick-Davies deduce exotic spheres in every dimension congruent to 72, 144, and 168 modulo 192 under their stated construction.

Recent work on free S^1 and S^3 actions on very exotic spheres reduces geometric classification to explicit products in $\pi_* S$, and some cases remain ambiguous precisely because those products are not known. This is a concrete example where the multiplicative frontier, not the additive group, is the obstruction.

12.4 Vector fields, immersions, and bundle theory

Adams solved the vector-fields-on-spheres problem: the maximum number of everywhere linearly independent tangent vector fields on S^{n-1} is $\rho(n)-1$, where ρ is the Radon-Hurwitz function. The proof uses K -theory, Adams operations, and the same height-one structure that computes $\text{image } J$.

Stable stems and J also enter immersion theory through normal invariants and obstruction groups. Pontryagin-Thom translates immersions and framed submanifolds into stable maps; surgery theory then uses stable normal data and L -groups to classify manifolds. Homotopy spheres are the most concentrated instance of this interaction.

12.5 Kervaire manifolds and framed geometry

A Kervaire-invariant-one class is represented by a framed manifold whose middle-dimensional mod-2 intersection form has Arf invariant one. The existence dimensions therefore encode exceptional framed manifolds, not merely spectral-sequence survivors. Hill-Hopkins-Ravenel's nonexistence theorem is a geometric theorem proved through equivariant stable homotopy; the proposed dimension-126 existence proof returns to detailed classical/synthetic Adams computation.

13. Algorithms, software, datasets, and machine proofs

13.1 Computability versus feasible computation

Brown proved that Postnikov complexes and higher homotopy groups of finite simply connected complexes are algorithmically computable. This positive theorem contrasts with the undecidability of fundamental-group triviality for arbitrary finite complexes. It does not make sphere groups easy: the resulting algorithms are far too expensive for the specialized high-stem calculations in this report.

For fixed homotopy degree, Cadek and collaborators give polynomial-time algorithms for homotopy groups and Postnikov systems of finite 1-connected simplicial sets. They also compute homotopy classes of maps into a sphere in a stable range. When the degree is part of the input, hardness results and representative-size lower bounds show severe complexity.

13.2 Effective homology and Kenzo

Effective homology replaces infinite chain complexes, such as loop spaces, with finite effective models connected by reductions. The Kenzo system implements iterated loop spaces, Eilenberg-Mac Lane spaces, spectral sequences, and Whitehead/Postnikov towers. It can compute examples inaccessible to naive simplicial enumeration.

Effective-homology output is complementary to classical named-generator tables. It may determine an abstract group for a specified finite space without identifying that group with Toda's conventional generators or resolving the global family structure.

13.3 Ext and chart computation

Bruner's minimal resolutions and the Bruner-Rognes database provide $\text{Ext}_A(F_2, F_2)$, products, and operations in large ranges. IWX publish classical and C-motivic Adams charts and algebraic Novikov/cofiber-of-tau charts on Zenodo. Guozhen Wang's morestablestems repository and related code expose machine-readable computations.

SeqSee introduces a JSON schema that separates spectral-sequence mathematics from visualization. This is an important reproducibility step: a chart should be generated from structured classes, products, and differentials rather than maintained as an opaque drawing.

13.4 Machine proofs among finite CW spectra

Lin-Wang-Xu's machine-proof project encodes finite spectra, maps, cofiber sequences, spectral-sequence naturality, and extension constraints. The v2 paper documents 49 CW spectra, 180 maps, and 61 cofiber sequences, while current metadata for the evolving dataset reports larger counts. The safe citation is therefore version-specific.

The machine does not replace mathematical input. Users specify cell complexes, maps, and trusted initial differentials; the solver propagates consequences and proves that alternatives are inconsistent. Human arguments still construct key complexes and identify which machine conclusion settles a homotopy question. The last Kervaire preprint is explicitly a hybrid of machine deductions and ad hoc insight.

13.5 Reproducibility requirements for future stem tables

1. Publish raw Ext and spectral-sequence objects, not only raster charts.
2. Version every differential and extension, with provenance and correction history.
3. Separate theorem-proved, machine-derived, assumed, and conjectural relations.
4. Record generator choices and indeterminacies for Toda brackets and hidden extensions.
5. Provide tests that recompute additive groups from the E_∞ page and extension data.
6. Export tables in a stable schema suitable for independent checking and formalization.

Homotopy computations are unusually vulnerable to transcription errors because one wrong differential changes several neighboring stems and products. The 2025 70-71 correction is a model for transparent repair: the authors identify the incorrect differential, state the corrected groups, and retract the associated hidden extension.

14. Current frontier and major open problems

Problem	Concrete target	Type
Complete stems 84-90	Resolve the explicit 2-primary alternatives and audit adjacent hidden extensions.	Finite computation
Extend the contiguous stable range	Compute full additive groups and products beyond 90.	Finite computation
Unstable 33-stem	Determine $\pi_{\{n+33\}}(S^n)_2$ for $10 \leq n \leq 27$ and clarify $n=27$.	Unstable/EHP
Higher unstable stems	Create complete all-sphere p-primary ranges beyond 33, ideally with machine-readable generators.	Unstable/EHP
Last Kervaire publication	Independent checking and peer-reviewed publication of the dimension-126 proof.	Foundational verification
Telescope counterexamples	Peer-reviewed consolidation and computation of the resulting difference between $T(n)$ - and $K(n)$ -local spheres.	Chromatic
Curtis conjecture	Classify spherical classes in $H_*(QS^0; F_2)$.	Hurewicz
Asymptotic growth	Determine whether stable-stem ranks have $\exp(\Theta((\log n)^3))$ growth or are much smaller.	Global
Vanishing stems	Find structural criteria or density results for $\pi_n S = 0$.	Global
Higher-height unit detection	Construct and classify v_n -periodic sphere families at heights $n \geq 3$.	Chromatic
Multiplication database	Audit the stable ring, Toda brackets, and power operations through the computed range.	Computational
Formal verification	Kernel-check the algebraic and finite-spectrum deduction layers without replacing expert mathematics by untrusted tables.	Foundations/software

14.1 Near-term numerical targets

The clearest finite project is an audited completion of stems 84-90 using synthetic Adams methods and the expanding finite-CW-spectrum machine database. Because the unresolved alternatives are explicit, success can be measured unambiguously. A second target is to push the exact stable range past 90 while retaining complete extension and product provenance.

Unstably, the 33-stem gap is similarly concrete. The challenge is not an unknown theory but a long chain of EHP, Toda bracket, and extension calculations. Modern machine assistance could check group-rank constraints and relation propagation, but the construction of unstable representatives remains substantially geometric.

14.2 Structural problems

At height two, the immediate task is to reconcile tmf -visible and tmf -invisible periodic families into a coherent description of the $K(2)$ -local and $T(2)$ -local unit. At higher heights, even the correct detectors and finite resolutions are under construction. The telescope counterexamples mean that one must specify which localization a family inhabits.

The Curtis conjecture, chromatic splitting phenomena, and asymptotic growth ask for qualitative control over infinitely many stems. They are harder than extending a table but may ultimately explain why the tables look as they do.

14.3 The role of formalization

The abstract foundations of spectra, generalized cohomology, and spectral sequences are increasingly formalizable, but current sphere computations depend on enormous external Ext datasets and thousands of named relations. A realistic formalization program should first verify data schemas, exact-couple propagation, finite-complex naturality, and extension reconstruction, while treating large Ext computations as separately certified inputs.

The most valuable initial benchmark would not be “formalize all stable stems through 90” as one monolith. It would be a layered certificate: raw chain complexes $\rightarrow$ Ext classes $\rightarrow$ spectral-sequence pages $\rightarrow$ differentials $\rightarrow$ E_∞ groups $\rightarrow$ hidden extensions $\rightarrow$ final additive groups, with each correction localized to a small certificate.

15. Chronology of major advances

Date	Authors	Advance
1931	Hopf	Hopf fibration and invariant.
1938	Freudenthal	Stable range under suspension.
1942-1951	Whitehead, Pontryagin, Rokhlin	J-homomorphism, framed geometry, first stable stems.
1951-1953	Serre	Spectral sequences, finiteness, rational groups, first p-torsion.
1953	Spanier-Whitehead	Stable homotopy category.
1955	James	Reduced products and EHP machinery.
1958-1962	Toda	p-primary methods, Toda brackets, complete 19-stem table.
1958-1966	Adams	Adams spectral sequence, Hopf invariant one, image J.
1959	Bott	Eightfold/twofold periodicity of classical groups.
1960-1969	Kervaire, Kervaire-Milnor, Browder	Kervaire invariant and exotic spheres.
1964	May	May spectral sequence.
1967-1969	Novikov, Quillen	Complex cobordism, Adams-Novikov, formal groups.
1973	Nishida	Nilpotence of positive stable classes.
1977	Miller-Ravenel-Wilson	Periodic Greek-letter families.
1978-1981	Kahn-Priddy, Lin, Mahowald	Transfer methods and bo-resolutions.
1979	Cohen-Moore-Neisendorfer; Oda	Odd-prime exponent theorem; unstable 25-31 stems at 2.
1984-1986	Ravenel and collaborators	Chromatic conjectures and synthesis in the green book.
1988-1998	Devinatz-Hopkins-Smith; Hopkins-Smith	Nilpotence, thick subcategories, periodicity.
1990s	Kochman, Mahowald, Bruner	Computer-assisted stable calculations.
1999-2001	Arone-Mahowald; Bousfield	Goodwillie and telescopic unstable homotopy.
2010-2019	Dugger-Isaksen, Isaksen, Gheorghe-Wang-Xu	Motivic deformation and stable stems through 59.
2016	Hill-Hopkins-Ravenel	Kervaire nonexistence above 126.
2017	Wang-Xu; Miyauchi-Mukai	61-stem vanishing; complete unstable 32-stem at 2.
2020-2023	Isaksen-Wang-Xu	Stable computation through 90 and smooth-structure tables.
2023-2026	BHLS; BIX; LWX; BBQ; Carrick-Davies; Yang-Wu	Telescope counterexamples, stem corrections, Kervaire preprint, new periodic families, unstable 33-stem progress.

16. Annotated bibliography and source map

The bibliography below is selective only in the narrow sense that it omits papers whose contribution is a single relation already indexed by a comprehensive source. It includes the foundational theorems, every principal numerical-range source, the modern computation pipeline, major periodic-family results, and the core geometric and algorithmic references. Entries are grouped by function rather than strict chronology.

16.1 Foundational unstable and stable topology

- Hans Freudenthal. “Über die Klassen der Sphärenabbildungen I. Große Dimensionen.” *Compos. Math.* 5, 299-314 (1938). *The suspension theorem establishing the stable range.*
- Heinz Hopf. “Über die Abbildungen der dreidimensionalen Sphäre auf die Kugelfläche.” *Math. Ann.* 104, 637-665 (1931). *Introduced the Hopf invariant and the Hopf fibration $S^3 \rightarrow S^2$, the starting point for low-dimensional sphere homotopy.*
- I. M. James. “Reduced product spaces.” *Ann. of Math.* 62, 170-197 (1955). *James construction and loop-space methods underlying the EHP sequence.*
- Lev S. Pontryagin. “Homotopy classification of mappings of an $(n+2)$ -dimensional sphere on an n -dimensional one.” *Dokl. Akad. Nauk SSSR* 70, 957-959 (1950). *Geometric framed-submanifold interpretation leading to Pontryagin-Thom theory.*
- V. A. Rokhlin. “The classification of mappings of the $(n+3)$ -sphere to the n -sphere.” *Dokl. Akad. Nauk SSSR* 81, 19-22 (1951). *Early geometric calculation of the third stable stem.*
- Jean-Pierre Serre. “Homologie singulière des espaces fibrés. Applications.” *Ann. of Math.* 54, 425-505 (1951). *Introduced spectral-sequence methods that produced the first substantial range of stable stems.*
- Jean-Pierre Serre. “Groupes d’homotopie et classes de groupes abéliens.” *Ann. of Math.* 58, 258-294 (1953). *Finiteness, rational structure, and first p -torsion results for homotopy groups of spheres.*
- Hiroshi Toda. “ p -primary components of homotopy groups. III. Stable groups of the sphere.” *Mem. Coll. Sci. Univ. Kyoto Ser. A Math.* 31, 191-210 (1958). *Classical odd-primary stable computations and early periodic families.*
- Hiroshi Toda. “Composition methods in homotopy groups of spheres.” *Annals of Mathematics Studies* 49, Princeton University Press (1962). *The definitive composition-method calculation of integral unstable groups through the 19-stem; introduced systematic Toda brackets.*
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- George W. Whitehead. “The $(n+2)$ nd homotopy group of the n -sphere.” *Ann. of Math.* 52, 245-247 (1950). *Early exact low-stem computation.*
- Edwin Spanier and George W. Whitehead. “A first approximation to homotopy theory.” *Proc. Natl. Acad. Sci. USA* 39, 655-660 (1953). *Foundational construction of the stable homotopy category.*

16.2 Complete unstable computation sources

- Mamoru Mimura. “On the generalized Hopf homomorphism and the higher composition, Part II: $\pi_{n+i}(S^n)$ for $i=21$ and 22 .” *J. Math. Kyoto Univ.* 4, 301-326 (1965). *Determined the 2-primary 21- and 22-stems.*
- Junichi Inoue and Juno Mukai. “The 31-stem homotopy groups of spheres.” *Kyushu J. Math.* / preprint literature (2012). *Bibliographic variants occur in secondary sources. Resolved remaining 2-primary 31-stem ambiguities.*
- Toshiyuki Miyauchi and Juno Mukai. “Determination of the 2-primary components of the 32-stem homotopy groups of S^n .” *Bol. Soc. Mat. Mex.* 23, 319-387 (2017). *Completed the 2-primary 32-stem for every $n \geq 2$ and determined $\pi_{42}(S^9)$ in the 33-stem.*
- Toshiyuki Miyauchi and Juno Mukai. “Relations in the 24-th homotopy groups of spheres.” *arXiv:2606.12046* (2026). *Proves nontrivial Toda brackets and relations needed for continuing unstable computations; not itself a complete new stem.*
- Mamoru Mimura, M. Mori, and Nobuyuki Oda. “Determinations of 2-components of the 23- and 24-stems in homotopy groups of spheres.” *Mem. Fac. Sci. Kyushu Univ.* 29, 1-42 (1975). *Completed the 2-primary 23- and 24-stems.*
- Nobuyuki Oda. “Unstable homotopy groups of spheres.” *Bull. Inst. Adv. Res. Fukuoka Univ.* 44, 49-152 (1979). *Massive determination of the 2-primary 25-31 stems for all spheres and substantial 32-33 stem data.*
- K. Oguchi. “Generators of 2-primary components of homotopy groups of spheres, unitary groups and symplectic groups.” *J. Fac. Sci. Univ. Tokyo* 11, 65-111 (1964). *Generator-level unstable 2-primary calculations used by later tables.*
- S. Thomeier. “Einige Ergebnisse über Homotopiegruppen von Sphären.” *Math. Ann.* 164, 225-250 (1966). *Gives the eight-step backward determination theorem used to recover near-stable unstable groups from stable stems.*
- Mamoru Mimura and Hiroshi Toda. “The $(n+20)$ -th homotopy groups of n -spheres.” *J. Math. Kyoto Univ.* 3, 37-58 (1963). *Completed the integral 20-stem.*
- Juxin Yang and Jie Wu. “On the extension problems for three 33-stem homotopy groups.” *Homology Homotopy Appl.* 28(1), 67-97 (2026). *Resolves $\pi_{39}(S^6)$, $\pi_{40}(S^7)$, and $\pi_{41}(S^8)$ at 2. The paper contains a range-caption inconsistency at $n=27$, documented in this report.*
- Guozhen Wang and Zhouli Xu. “A survey of computations of homotopy groups of spheres and cobordisms.” Preprint / survey (2010). *Detailed generator-level guide to classical stable and unstable computation literature.*

16.3 Adams, Adams-Novikov, and low stable stems

- J. Frank Adams. “On the structure and applications of the Steenrod algebra.” *Comment. Math. Helv.* 32, 180-214 (1958). *Introduced the Adams spectral sequence for stable homotopy calculations.*

- J. Frank Adams. “On the non-existence of elements of Hopf invariant one.” *Ann. of Math.* 72, 20-104 (1960). *Proved that the only Hopf-invariant-one dimensions are 1, 2, 4, and 8.*
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- Robert Bruner. “A new differential in the Adams spectral sequence.” *Topology* 23, 271-276 (1984). *A decisive differential in the classical 2-primary Adams spectral sequence.*
- Daniel Isaksen. “Stable stems.” *Mem. Amer. Math. Soc.* 262, no. 1269 (2019). *Motivic calculation of the 2-complete stable stems through 59 and supporting Adams/Adams-Novikov data.*
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- Douglas Ravenel. “Complex cobordism and stable homotopy groups of spheres.” *Pure and Applied Mathematics* 121, Academic Press (1986). *The standard reference for Adams-Novikov and chromatic calculations, including extensive $p=3$ and $p=5$ tables.*
- Mark Mahowald and Martin Tangora. “Some differentials in the Adams spectral sequence.” *Topology* 6, 349-369 (1967). *Central classical differential computations extending stable-stem ranges.*
- Michael Barratt, Mark Mahowald, and Martin Tangora. “Some differentials in the Adams spectral sequence. II.” *Topology* 9, 309-316 (1970). *Further Adams differentials and low-stem calculations.*
- Martin Tangora. “Some homotopy groups mod 3.” *Conference on Homotopy Theory, Notas Mat. Simpos.* 1, 227-245 (1975). *Classical 3-primary Adams computations.*
- Shichiro Oka and Hiroshi Toda. “3-primary beta-family in stable homotopy.” *Hiroshima Math. J.* 5, 447-460 (1975). *Constructs and analyzes the 3-primary beta family.*
- Guozhen Wang and Zhouli Xu. “The triviality of the 61-stem in the stable homotopy groups of spheres.” *Ann. of Math.* 186, 501-580 (2017). *Proves the 2-primary 61-stem vanishes and derives the uniqueness of the smooth 61-sphere.*
- Daniel Isaksen, Guozhen Wang, and Zhouli Xu. “Stable homotopy groups of spheres.” *Proc. Natl. Acad. Sci. USA* 117, 24757-24763 (2020). *Concise overview of motivic computation and smooth-structure applications.*
- Guozhen Wang and Zhouli Xu. “Stable homotopy groups.” *EMS Magazine* 128, 23-29 (2023). *Compact modern survey recording the classical $p=3$ range through 108 and $p=5$ range through 999.*
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- Robert Burklund, Daniel Isaksen, and Zhouli Xu. “Classical stable homotopy groups of spheres via F_2 -synthetic methods.” *Peking Math. J.*, published 11 April 2025 (2025). *Completes stems 82 and 83 and corrects the 2-primary v_1 -torsion in stems 70 and 71.*

16.4 Hopf, J, Kervaire, and smooth structures

- J. Frank Adams. “Vector fields on spheres.” *Ann. of Math.* 75, 603-632 (1962). *Solves the vector-fields-on-spheres problem using K-theory and Hopf-invariant methods.*
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- Michel Kervaire and John Milnor. “Groups of homotopy spheres. I.” *Ann. of Math.* 77, 504-537 (1963). *Relates exotic spheres to stable stems, coker J , parallelizable boundaries, and the Kervaire invariant.*

- Daniel Quillen. “The Adams conjecture.” *Topology* 10, 67-80 (1971). *Proof of the Adams conjecture and hence the image-of-J calculation.*
- Michael Hill, Michael Hopkins, and Douglas Ravenel. “On the nonexistence of elements of Kervaire invariant one.” *Ann. of Math.* 184, 1-262 (2016). *Rules out Kervaire-invariant-one elements in dimensions above 126.*
- Michael Atiyah, Raoul Bott, and Arnold Shapiro. “Clifford modules.” *Topology* 3, Suppl. 1, 3-38 (1964). *K-theoretic and Clifford-algebra framework for vector fields and image J.*
- Stephen Smale. “Generalized Poincaré’s conjecture in dimensions greater than four.” *Ann. of Math.* 74, 391-406 (1961). *High-dimensional h-cobordism and Poincaré theory, the geometric setting for homotopy spheres.*
- Dennis Sullivan. “Genetics of homotopy theory and the Adams conjecture.” *Ann. of Math.* 100, 1-79 (1974). *Independent proof and localization-theoretic framework for the Adams conjecture.*
- Weinan Lin, Guozhen Wang, and Zhouli Xu. “On the Last Kervaire Invariant Problem.” arXiv:2412.10879v2 (2025). Preprint as of 8 August 2026. *Claims survival of $h_{-6}^{\wedge 2}$ and existence in dimension 126, completing the dimension list 2,6,14,30,62,126 when combined with earlier theorems.*
- Tilman Bauer and collaborators. “Infinite families of very exotic spheres with free $S^{\wedge 1}$ - and $S^{\wedge 3}$ -actions.” arXiv:2603.23241 (2026). *A recent geometric application whose residual ambiguities explicitly depend on stable-stem products.*

16.5 Chromatic structure and periodic families

- A. K. Bousfield. “On the homotopy theory of K-local spectra at an odd prime.” *Amer. J. Math.* 107 / related series (1982). See Bousfield’s K-local papers for exact bibliographic segmentation. *Computes the height-one local category and v_1 -periodic information.*
- Christian Carrick and Jack Morgan Davies. “A synthetic approach to detecting v_1 -periodic families.” arXiv:2401.16508; accepted *Trans. AMS* (2024). *Shows the unit to the connective image-of-J spectrum detects all v_1 -periodic classes in the stated sense.*
- Christian Carrick and Jack Morgan Davies. “Nonvanishing of products in v_2 -periodic families at the prime 3.” arXiv:2410.02564 (2024). *Establishes extensive nonzero products among 3-primary v_2 -periodic families.*
- Christian Carrick and Jack Morgan Davies. “On periodic families in the stable stems of height two.” arXiv:2506.20507 (2025). *Finds families in 19 residue classes modulo 192, including simple 4- and 8-torsion, and derives exotic-sphere consequences.*
- Jack Morgan Davies. “Revisiting the β_1 -action on the 3-primary stable homotopy groups of spheres.” arXiv:2511.02324 (2025). *Proves $\beta_1^{\wedge 5}$ is nonzero and $\beta_1^{\wedge 6}$ is zero and extends periodic product calculations.*
- Runji Li and Yuxuan Li. “The Adams differentials on the e-family.” arXiv:2602.20184; *Proc. Amer. Math. Soc.* (2026). *Proves the New Doomsday Conjecture for the e-family on the Adams 4-line.*
- Haynes Miller. “On relations between Adams spectral sequences, with an application to the stable homotopy of a Moore space.” *J. Pure Appl. Algebra* 20, 287-312 (1981). *Foundational change-of-rings and height-one/telescopic techniques.*
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- Irina Bobkova and J. D. Quigley. “New simple eta-torsion families of elements in the stable stems.” arXiv:2410.21181 (2024). *Five new 192-periodic simple eta-torsion families with trivial tmf -Hurewicz image.*
- Prasit Bhattacharya, Irina Bobkova, and J. D. Quigley. “New infinite families in the stable homotopy groups of spheres.” *Geom. Topol.* 30(6) / arXiv:2404.10062 (2026). *Seven new 192-periodic 2-primary families invisible to tmf but nonzero after $T(2)$ - and $K(2)$ -localization.*
- Douglas Ravenel. “Localization with respect to certain periodic homology theories.” *Amer. J. Math.* 106, 351-414 (1984). *Formulates the chromatic and telescope conjectures and develops localization methods.*
- Robert Burklund, Jeremy Hahn, Ishan Levy, and Tomer Schlank. “K-theoretic counterexamples to Ravenel’s telescope conjecture.” arXiv:2310.17459 (2023). Preprint as of 8 August 2026. *Claims the telescope conjecture fails at every prime and every chromatic height at least 2.*
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- Michael Hopkins and Jeffrey Smith. “Nilpotence and stable homotopy theory. II.” *Ann. of Math.* 148, 1-49 (1998). *Thick subcategory classification and periodicity theorem for finite p-local spectra.*
- Haynes Miller, Douglas Ravenel, and W. Stephen Wilson. “Periodic phenomena in the Adams-Novikov spectral sequence.” *Ann. of Math.* 106, 469-516 (1977). *Constructs Greek-letter families and establishes large-scale chromatic periodicity phenomena.*
- Agnes Beaudry, Mark Behrens, Prasit Bhattacharya, Dominic Culver, and Zhouli Xu. “The telescope conjecture at height 2 and the tmf resolution.” *J. Topol.* 14, 1243-1320 (2021). *Computational framework for height-two telescope questions using tmf .*
- Robert Burklund and Zhouli Xu. “The Adams differentials on the classes $h_j^{\wedge 3}$.” *Invent. Math.* 239, 1-77 (2025). *Determines an infinite d_4 -family and controls $h_j^{\wedge 2}$ through later Adams pages, feeding the Kervaire analysis.*

16.6 Unstable chromatic and global structure

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- Gregory Arone and Mark Mahowald. “The Goodwillie tower of the identity functor and the unstable periodic homotopy of spheres.” *Invent. Math.* 135, 743-788 (1999). *Relates Goodwillie layers to v_n -periodic unstable homotopy of spheres.*
- Mark Behrens and Connor Malin. “Unstable homotopy groups and Lie algebras.” *arXiv:2410.18330* (2024). *Modern Lie-algebraic organization of unstable chromatic homotopy.*
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- Robert Burklund, with appendix by Andrew Senger. “How big are the stable homotopy groups of spheres?.” *arXiv:2203.00670* (2022). *Proves sublinear p -torsion exponents and subexponential rank/size upper bounds; includes unstable bounds.*
- Robert Wellington. “The unstable Adams spectral sequence for free iterated loop spaces.” *Mem. Amer. Math. Soc.* 36, no. 258 (1982). *Major calculations and evidence concerning spherical classes and unstable Adams methods.*
- Roman Mikhailov and Jie Wu. “Homotopy groups as centers of finitely presented groups.” *Algebr. Geom. Topol. / related papers* (2010). *Realizes sphere homotopy groups through centers of explicit combinatorial groups.*
- Sergei Ivanov, Roman Mikhailov, and Jie Wu. “On nontriviality of homotopy groups of spheres.” *arXiv:1506.00952 / related publication* (2017). *Proves $\pi_n(S^{\wedge 2})$ is nonzero for every $n \geq 2$, hence $\pi_n(S^{\wedge 3})$ is nonzero for every $n \geq 3$.*

16.7 Motivic, synthetic, and computational infrastructure

- Daniel Dugger and Daniel Isaksen. “The motivic Adams spectral sequence.” *Geom. Topol.* 14, 967-1014 (2010). *Computational motivic Adams machinery over C .*
- Joey Beauvais-Feisthauer and Daniel Isaksen. “SeqSee: a schema-based approach to spectral sequence visualization.” *arXiv:2501.18429v2* (2025). *Standardized JSON representation and visualization system for Adams-type spectral sequences.*
- Edgar H. Brown Jr.. “Finite computability of Postnikov complexes.” *Ann. of Math.* 65, 1-20 (1957). *Proves higher homotopy groups of finite simply connected complexes are algorithmically computable.*
- Piotr Pstragowski. “Synthetic spectra and the cellular motivic category.” *Invent. Math. / published version of arXiv:1803.01804* (2023). *Constructs synthetic spectra, a flexible deformation framework used in the 2025 stem corrections.*
- Robert Bruner and John Rognes. “The cohomology of the mod 2 Steenrod algebra [data set].” *Norstore*, DOI 10.11582/2021.00077 (2021). *Machine-readable Ext database supporting Adams calculations.*
- Julio Rubio and Francis Sergeraert. “Constructive algebraic topology and effective homology.” *Kenzo/effective-homology literature* (1990s-2010s). *Practical algorithms for iterated loop spaces, Postnikov/Whitehead towers, and homotopy computations.*
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- Martin Cadek, Marek Krcal, Jiri Matousek, Lukas Vokrinek, and Uli Wagner. “Polynomial-time computation of homotopy groups and Postnikov systems in fixed dimension.” *J. ACM / arXiv:1211.3093* (2018). *Polynomial-time algorithms when the homotopy degree is fixed.*
- Daniel Isaksen and Zhouli Xu. “Motivic stable homotopy and the stable 51 and 52 stems.” *Topology Appl.* 190, 31-34 (2015). *Early demonstration that motivic information resolves classical stable-stem extensions.*
- Bogdan Gheorghe, Guozhen Wang, and Zhouli Xu. “The special fiber of the motivic deformation of the stable homotopy category is algebraic.” *Acta Math.* 226, 319-407 (2021). *Identifies the cofiber-of-tau category with an algebraic model, central to modern computations.*
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- Weinan Lin, Guozhen Wang, and Zhouli Xu. “Machine proofs for Adams differentials and extension problems among CW spectra.” *arXiv:2412.10876v2* and *Zenodo dataset* (2025). *Documents reproducible machine deductions among 49 spectra, 180 maps, and 61 cofiber sequences in v_2 ; newer metadata pages report a larger evolving dataset.*
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Appendix A. Stable homotopy groups of spheres, stems 0-90

Notation: “ 2^r ” in the compact source column means r copies of $\mathbb{Z}/2$, products separated by dots in IWX are written as direct sums, and alternatives separated by OR are unresolved. The decomposition is into 2-primary v_1 -torsion, odd-primary v_1 -torsion, and v_1 -periodic subgroups. Stems 70, 71, 82, and 83 incorporate the 2025 corrections/completions.

k	2-primary v_1 -torsion	Odd v_1 -torsion	v_1 -periodic	$\pi_k S$ (additive)	Status
0	$\mathbb{Z}$	0	0	$\mathbb{Z}$	exact
1	0	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	exact
2	0	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	exact
3	0	0	$\mathbb{Z}/8 + \mathbb{Z}/3$	$\mathbb{Z}/8 + \mathbb{Z}/3$	exact
4	0	0	0	0	exact
5	0	0	0	0	exact
6	$\mathbb{Z}/2$	0	0	$\mathbb{Z}/2$	exact
7	0	0	$\mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/5$	$\mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/5$	exact
8	$\mathbb{Z}/2$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2 + \mathbb{Z}/2$	exact
9	$\mathbb{Z}/2$	0	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/2 + (\mathbb{Z}/2)^2$	exact
10	0	$\mathbb{Z}/3$	$\mathbb{Z}/2$	$\mathbb{Z}/3 + \mathbb{Z}/2$	exact
11	0	0	$\mathbb{Z}/8 + \mathbb{Z}/9 + \mathbb{Z}/7$	$\mathbb{Z}/8 + \mathbb{Z}/9 + \mathbb{Z}/7$	exact
12	0	0	0	0	exact
13	0	$\mathbb{Z}/3$	0	$\mathbb{Z}/3$	exact
14	$\mathbb{Z}/2 + \mathbb{Z}/2$	0	0	$\mathbb{Z}/2 + \mathbb{Z}/2$	exact
15	$\mathbb{Z}/2$	0	$\mathbb{Z}/32 + \mathbb{Z}/3 + \mathbb{Z}/5$	$\mathbb{Z}/2 + \mathbb{Z}/32 + \mathbb{Z}/3 + \mathbb{Z}/5$	exact
16	$\mathbb{Z}/2$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2 + \mathbb{Z}/2$	exact
17	$(\mathbb{Z}/2)^2$	0	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^2 + (\mathbb{Z}/2)^2$	exact
18	$\mathbb{Z}/8$	0	$\mathbb{Z}/2$	$\mathbb{Z}/8 + \mathbb{Z}/2$	exact
19	$\mathbb{Z}/2$	0	$\mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/11$	$\mathbb{Z}/2 + \mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/11$	exact
20	$\mathbb{Z}/8$	$\mathbb{Z}/3$	0	$\mathbb{Z}/8 + \mathbb{Z}/3$	exact
21	$(\mathbb{Z}/2)^2$	0	0	$(\mathbb{Z}/2)^2$	exact
22	$(\mathbb{Z}/2)^2$	0	0	$(\mathbb{Z}/2)^2$	exact
23	$\mathbb{Z}/2 + \mathbb{Z}/8$	$\mathbb{Z}/3$	$\mathbb{Z}/16 + \mathbb{Z}/9 + \mathbb{Z}/5 + \mathbb{Z}/7 + \mathbb{Z}/13$	$\mathbb{Z}/2 + \mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/16 + \mathbb{Z}/9 + \mathbb{Z}/5 + \mathbb{Z}/7 + \mathbb{Z}/13$	exact
24	$\mathbb{Z}/2$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2 + \mathbb{Z}/2$	exact
25	0	0	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^2$	exact
26	$\mathbb{Z}/2$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	$\mathbb{Z}/2 + \mathbb{Z}/3 + \mathbb{Z}/2$	exact
27	0	0	$\mathbb{Z}/8 + \mathbb{Z}/3$	$\mathbb{Z}/8 + \mathbb{Z}/3$	exact
28	$\mathbb{Z}/2$	0	0	$\mathbb{Z}/2$	exact
29	0	$\mathbb{Z}/3$	0	$\mathbb{Z}/3$	exact
30	$\mathbb{Z}/2$	$\mathbb{Z}/3$	0	$\mathbb{Z}/2 + \mathbb{Z}/3$	exact
31	$(\mathbb{Z}/2)^2$	0	$\mathbb{Z}/64 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/17$	$(\mathbb{Z}/2)^2 + \mathbb{Z}/64 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/17$	exact
32	$(\mathbb{Z}/2)^3$	0	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^3 + \mathbb{Z}/2$	exact
33	$(\mathbb{Z}/2)^3$	0	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^3 + (\mathbb{Z}/2)^2$	exact
34	$(\mathbb{Z}/2)^2 + \mathbb{Z}/4$	0	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^2 + \mathbb{Z}/4 + \mathbb{Z}/2$	exact

k	2-primary v1-torsion	Odd v1-torsion	v1-periodic	π_k^S (additive)	Status
35	$(\mathbb{Z}/2)^2$	0	$\mathbb{Z}/8 + \mathbb{Z}/27 + \mathbb{Z}/7 + \mathbb{Z}/19$	$(\mathbb{Z}/2)^2 + \mathbb{Z}/8 + \mathbb{Z}/27 + \mathbb{Z}/7 + \mathbb{Z}/19$	exact
36	$\mathbb{Z}/2$	$\mathbb{Z}/3$	0	$\mathbb{Z}/2 + \mathbb{Z}/3$	exact
37	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/3$	0	$(\mathbb{Z}/2)^2 + \mathbb{Z}/3$	exact
38	$\mathbb{Z}/2 + \mathbb{Z}/4$	$\mathbb{Z}/3 + \mathbb{Z}/5$	0	$\mathbb{Z}/2 + \mathbb{Z}/4 + \mathbb{Z}/3 + \mathbb{Z}/5$	exact
39	$(\mathbb{Z}/2)^5$	$\mathbb{Z}/3$	$\mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/25 + \mathbb{Z}/11$	$(\mathbb{Z}/2)^5 + \mathbb{Z}/3 + \mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/25 + \mathbb{Z}/11$	exact
40	$(\mathbb{Z}/2)^4 + \mathbb{Z}/4$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^4 + \mathbb{Z}/4 + \mathbb{Z}/3 + \mathbb{Z}/2$	exact
41	$(\mathbb{Z}/2)^3$	0	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^3 + (\mathbb{Z}/2)^2$	exact
42	$\mathbb{Z}/2 + \mathbb{Z}/8$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	$\mathbb{Z}/2 + \mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/2$	exact
43	0	0	$\mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/23$	$\mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/23$	exact
44	$\mathbb{Z}/8$	0	0	$\mathbb{Z}/8$	exact
45	$(\mathbb{Z}/2)^3 + \mathbb{Z}/16$	$\mathbb{Z}/9 + \mathbb{Z}/5$	0	$(\mathbb{Z}/2)^3 + \mathbb{Z}/16 + \mathbb{Z}/9 + \mathbb{Z}/5$	exact
46	$(\mathbb{Z}/2)^4$	$\mathbb{Z}/3$	0	$(\mathbb{Z}/2)^4 + \mathbb{Z}/3$	exact
47	$(\mathbb{Z}/2)^3 + \mathbb{Z}/4$	$\mathbb{Z}/3$	$\mathbb{Z}/32 + \mathbb{Z}/9 + \mathbb{Z}/5 + \mathbb{Z}/7 + \mathbb{Z}/13$	$(\mathbb{Z}/2)^3 + \mathbb{Z}/4 + \mathbb{Z}/3 + \mathbb{Z}/32 + \mathbb{Z}/9 + \mathbb{Z}/5 + \mathbb{Z}/7 + \mathbb{Z}/13$	exact
48	$(\mathbb{Z}/2)^3 + \mathbb{Z}/4$	0	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^3 + \mathbb{Z}/4 + \mathbb{Z}/2$	exact
49	0	$\mathbb{Z}/3$	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/3 + (\mathbb{Z}/2)^2$	exact
50	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^2 + \mathbb{Z}/3 + \mathbb{Z}/2$	exact
51	$\mathbb{Z}/2 + \mathbb{Z}/8$	0	$\mathbb{Z}/8 + \mathbb{Z}/3$	$\mathbb{Z}/2 + \mathbb{Z}/8 + \mathbb{Z}/8 + \mathbb{Z}/3$	exact
52	$(\mathbb{Z}/2)^3$	$\mathbb{Z}/3$	0	$(\mathbb{Z}/2)^3 + \mathbb{Z}/3$	exact
53	$(\mathbb{Z}/2)^4$	0	0	$(\mathbb{Z}/2)^4$	exact
54	$\mathbb{Z}/2 + \mathbb{Z}/4$	0	0	$\mathbb{Z}/2 + \mathbb{Z}/4$	exact
55	0	$\mathbb{Z}/3$	$\mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/29$	$\mathbb{Z}/3 + \mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/29$	exact
56	0	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	exact
57	$\mathbb{Z}/2$	0	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/2 + (\mathbb{Z}/2)^2$	exact
58	$\mathbb{Z}/2$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2 + \mathbb{Z}/2$	exact
59	$(\mathbb{Z}/2)^2$	0	$\mathbb{Z}/8 + \mathbb{Z}/9 + \mathbb{Z}/7 + \mathbb{Z}/11 + \mathbb{Z}/31$	$(\mathbb{Z}/2)^2 + \mathbb{Z}/8 + \mathbb{Z}/9 + \mathbb{Z}/7 + \mathbb{Z}/11 + \mathbb{Z}/31$	exact
60	$\mathbb{Z}/4$	0	0	$\mathbb{Z}/4$	exact
61	0	0	0	0	exact
62	$(\mathbb{Z}/2)^4$	$\mathbb{Z}/3$	0	$(\mathbb{Z}/2)^4 + \mathbb{Z}/3$	exact
63	$(\mathbb{Z}/2)^2 + \mathbb{Z}/4$	0	$\mathbb{Z}/128 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/17$	$(\mathbb{Z}/2)^2 + \mathbb{Z}/4 + \mathbb{Z}/128 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/17$	exact
64	$(\mathbb{Z}/2)^5 + \mathbb{Z}/4$	0	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^5 + \mathbb{Z}/4 + \mathbb{Z}/2$	exact
65	$(\mathbb{Z}/2)^7 + \mathbb{Z}/4$	$\mathbb{Z}/3$	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^7 + \mathbb{Z}/4 + \mathbb{Z}/3 + (\mathbb{Z}/2)^2$	exact
66	$(\mathbb{Z}/2)^5 + \mathbb{Z}/8$	0	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^5 + \mathbb{Z}/8 + \mathbb{Z}/2$	exact
67	$(\mathbb{Z}/2)^3 + \mathbb{Z}/4$	0	$\mathbb{Z}/8 + \mathbb{Z}/3$	$(\mathbb{Z}/2)^3 + \mathbb{Z}/4 + \mathbb{Z}/8 + \mathbb{Z}/3$	exact
68	$(\mathbb{Z}/2)^3$	$\mathbb{Z}/3$	0	$(\mathbb{Z}/2)^3 + \mathbb{Z}/3$	exact
69	$(\mathbb{Z}/2)^4$	0	0	$(\mathbb{Z}/2)^4$	exact
70	$(\mathbb{Z}/2)^6 + \mathbb{Z}/4$	0	0	$(\mathbb{Z}/2)^6 + \mathbb{Z}/4$	exact; corrected 2025
71	$(\mathbb{Z}/2)^5 + \mathbb{Z}/4 + \mathbb{Z}/8$	0	$\mathbb{Z}/16 + \mathbb{Z}/27 + \mathbb{Z}/5 + \mathbb{Z}/7 + \mathbb{Z}/13 + \mathbb{Z}/19 + \mathbb{Z}/37$	$(\mathbb{Z}/2)^5 + \mathbb{Z}/4 + \mathbb{Z}/8 + \mathbb{Z}/16 + \mathbb{Z}/27 + \mathbb{Z}/5 + \mathbb{Z}/7 + \mathbb{Z}/13 + \mathbb{Z}/19 + \mathbb{Z}/37$	exact; corrected 2025
72	$(\mathbb{Z}/2)^7$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^7 + \mathbb{Z}/3 + \mathbb{Z}/2$	exact

k	2-primary v1-torsion	Odd v1-torsion	v1-periodic	π_k^S (additive)	Status
73	$(\mathbb{Z}/2)^{\wedge 5}$	0	$(\mathbb{Z}/2)^{\wedge 2}$	$(\mathbb{Z}/2)^{\wedge 5} + (\mathbb{Z}/2)^{\wedge 2}$	exact
74	$(\mathbb{Z}/4)^{\wedge 3}$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	$(\mathbb{Z}/4)^{\wedge 3} + \mathbb{Z}/3 + \mathbb{Z}/2$	exact
75	$\mathbb{Z}/2$	$\mathbb{Z}/9$	$\mathbb{Z}/8 + \mathbb{Z}/3$	$\mathbb{Z}/2 + \mathbb{Z}/9 + \mathbb{Z}/8 + \mathbb{Z}/3$	exact
76	$(\mathbb{Z}/2)^{\wedge 2} + \mathbb{Z}/4$	$\mathbb{Z}/5$	0	$(\mathbb{Z}/2)^{\wedge 2} + \mathbb{Z}/4 + \mathbb{Z}/5$	exact
77	$(\mathbb{Z}/2)^{\wedge 5} + \mathbb{Z}/4$	0	0	$(\mathbb{Z}/2)^{\wedge 5} + \mathbb{Z}/4$	exact
78	$(\mathbb{Z}/2)^{\wedge 3} + (\mathbb{Z}/4)^{\wedge 2}$	$\mathbb{Z}/3$	0	$(\mathbb{Z}/2)^{\wedge 3} + (\mathbb{Z}/4)^{\wedge 2} + \mathbb{Z}/3$	exact
79	$(\mathbb{Z}/2)^{\wedge 6} + \mathbb{Z}/4$	0	$\mathbb{Z}/32 + \mathbb{Z}/3 + \mathbb{Z}/25 + \mathbb{Z}/11 + \mathbb{Z}/41$	$(\mathbb{Z}/2)^{\wedge 6} + \mathbb{Z}/4 + \mathbb{Z}/32 + \mathbb{Z}/3 + \mathbb{Z}/25 + \mathbb{Z}/11 + \mathbb{Z}/41$	exact
80	$(\mathbb{Z}/2)^{\wedge 8}$	0	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^{\wedge 8} + \mathbb{Z}/2$	exact
81	$(\mathbb{Z}/2)^{\wedge 3} + \mathbb{Z}/4 + \mathbb{Z}/8$	$(\mathbb{Z}/3)^{\wedge 2}$	$(\mathbb{Z}/2)^{\wedge 2}$	$(\mathbb{Z}/2)^{\wedge 3} + \mathbb{Z}/4 + \mathbb{Z}/8 + (\mathbb{Z}/3)^{\wedge 2} + (\mathbb{Z}/2)^{\wedge 2}$	exact
82	$(\mathbb{Z}/2)^{\wedge 5} + \mathbb{Z}/8$	$\mathbb{Z}/3 + \mathbb{Z}/7$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^{\wedge 5} + \mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/7 + \mathbb{Z}/2$	exact; completed 2025
83	$(\mathbb{Z}/2)^{\wedge 3} + \mathbb{Z}/8$	$\mathbb{Z}/5$	$\mathbb{Z}/8 + \mathbb{Z}/9 + \mathbb{Z}/49 + \mathbb{Z}/43$	$(\mathbb{Z}/2)^{\wedge 3} + \mathbb{Z}/8 + \mathbb{Z}/5 + \mathbb{Z}/8 + \mathbb{Z}/9 + \mathbb{Z}/49 + \mathbb{Z}/43$	exact; completed 2025
84	$(\mathbb{Z}/2)^{\wedge 6}$ OR $(\mathbb{Z}/2)^{\wedge 5}$	$(\mathbb{Z}/3)^{\wedge 2}$	0	$[(\mathbb{Z}/2)^{\wedge 6} + (\mathbb{Z}/3)^{\wedge 2}]$ OR $[(\mathbb{Z}/2)^{\wedge 5} + (\mathbb{Z}/3)^{\wedge 2}]$	partial: one unresolved $\mathbb{Z}/2$
85	$(\mathbb{Z}/2)^{\wedge 6} + (\mathbb{Z}/4)^{\wedge 2}$ OR $(\mathbb{Z}/2)^{\wedge 5} + (\mathbb{Z}/4)^{\wedge 2}$ OR $(\mathbb{Z}/2)^{\wedge 4} + (\mathbb{Z}/4)^{\wedge 3}$ OR $(\mathbb{Z}/2)^{\wedge 7} + \mathbb{Z}/4$	$(\mathbb{Z}/3)^{\wedge 2}$	0	$[(\mathbb{Z}/2)^{\wedge 6} + (\mathbb{Z}/4)^{\wedge 2} + (\mathbb{Z}/3)^{\wedge 2}]$ OR $[(\mathbb{Z}/2)^{\wedge 5} + (\mathbb{Z}/4)^{\wedge 2} + (\mathbb{Z}/3)^{\wedge 2}]$ OR $[(\mathbb{Z}/2)^{\wedge 4} + (\mathbb{Z}/4)^{\wedge 3} + (\mathbb{Z}/3)^{\wedge 2}]$ OR $[(\mathbb{Z}/2)^{\wedge 7} + \mathbb{Z}/4 + (\mathbb{Z}/3)^{\wedge 2}]$	partial: extension/differential ambiguity
86	$(\mathbb{Z}/2)^{\wedge 4} + (\mathbb{Z}/8)^{\wedge 2}$ OR $(\mathbb{Z}/2)^{\wedge 2} + \mathbb{Z}/4 + (\mathbb{Z}/8)^{\wedge 2}$	$\mathbb{Z}/3 + \mathbb{Z}/5$	0	$[(\mathbb{Z}/2)^{\wedge 4} + (\mathbb{Z}/8)^{\wedge 2} + \mathbb{Z}/3 + \mathbb{Z}/5]$ OR $[(\mathbb{Z}/2)^{\wedge 2} + \mathbb{Z}/4 + (\mathbb{Z}/8)^{\wedge 2} + \mathbb{Z}/3 + \mathbb{Z}/5]$	partial: 2-primary ambiguity
87	$(\mathbb{Z}/2)^{\wedge 5} + \mathbb{Z}/4$	0	$\mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/23$	$(\mathbb{Z}/2)^{\wedge 5} + \mathbb{Z}/4 + \mathbb{Z}/16 + \mathbb{Z}/3 + \mathbb{Z}/5 + \mathbb{Z}/23$	exact
88	$(\mathbb{Z}/2)^{\wedge 4} + \mathbb{Z}/4$	0	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^{\wedge 4} + \mathbb{Z}/4 + \mathbb{Z}/2$	exact
89	$(\mathbb{Z}/2)^{\wedge 3}$	0	$(\mathbb{Z}/2)^{\wedge 2}$	$(\mathbb{Z}/2)^{\wedge 3} + (\mathbb{Z}/2)^{\wedge 2}$	exact
90	$(\mathbb{Z}/2)^{\wedge 3} + \mathbb{Z}/8$ OR $(\mathbb{Z}/2)^{\wedge 2} + \mathbb{Z}/8$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	$[(\mathbb{Z}/2)^{\wedge 3} + \mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/2]$ OR $[(\mathbb{Z}/2)^{\wedge 2} + \mathbb{Z}/8 + \mathbb{Z}/3 + \mathbb{Z}/2]$	partial: one unresolved $\mathbb{Z}/2$

Table A1. Corrected stable additive-group table through stem 90. Source: IWX 2023, updated by Burklund-Isaksen-Xu 2025.

Appendix B. Toda's complete integral unstable table through the 19-stem

The table is split into two panels for legibility. Rows are offsets k ; columns are sphere dimensions n ; the entry is $\pi_{\{n+k\}}(S^n)$. Compact notation: $1=0$, $\infty=\mathbb{Z}$, “+”=direct sum, and a^r means r copies of $\mathbb{Z}/a$. The final column gives the stable group.

k	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$	$n=9$	$n=10$
0	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
1	1	∞	2	2	2	2	2	2	2	2
2	1	2	2	2	2	2	2	2	2	2
3	1	2	$4+3$	$\infty+4+3$	$8+3$	$8+3$	$8+3$	$8+3$	$8+3$	$8+3$
4	1	$4+3$	2	2^2	2	1	1	1	1	1
5	1	2	2	2^2	2	∞	1	1	1	1
6	1	2	3	$8+3+3$	2	2	2	2	2	2
7	1	3	$3+5$	$3+5$	$2+3+5$	$4+3+5$	$8+3+5$	$\infty+8+3+5$	$16+3+5$	$16+3+5$
8	1	$3+5$	2	2	2	$8+2+3$	2^3	2^4	2^2	2^2
9	1	2	2^2	2^3	2^3	2^3	2^4	2^5	2^4	$\infty+2^3$
10	1	2^2	$4+2+3$	$8+4+2+3^2+5$	$8+2+9$	$8+2+9$	$8+3+2$	8^2+2+3^2	$8+2+3$	$4+2+3$
11	1	$4+2+3$	$4+2^2+3+7$	$4+2^5+3+7$	$8+2^2+9+7$	$8+4+9+7$	$8+2+9+7$	$8+2+9+7$	$8+2+9+7$	$8+2+9+7$
12	1	$4+2^2+3+7$	2^2	2^6	2^3	$16+3+5$	1	1	1	$4+3$
13	1	2^2	$2+3$	$8+2^2+3^2$	2^2+3	$2+3$	$2+3$	2^2+3	$2+3$	$2+3$
14	1	$2+3$	$2+3+5$	$8+2^2+9+3+5+7$	2^2+3	$4+2+3$	$8+4+3$	$16+8+4+3^2+5$	$16+4$	$16+2$
15	1	$2+3+5$	$2+3+5$	$2+3+5$	2^2+3+5	$4+2+3^2+5$	$8+2^3+3+5$	$8+2^5+3+5$	$16+2^3+3+5$	$16+2^2+3+5$
16	1	$2+3+5$	2^2+3	2^3+3^2	2^2	$8+2^2+9+7$	2^4	2^7	2^4	$16+2+3+5$
17	1	2^2+3	$4+2^2+3$	$8+4^2+2^2+3^2$	$4+2^2$	2^4	2^4	2^5+3	2^4	2^3
18	1	$4+2^2+3$	$4+2^2+3$	$8+4+2^5+3^2+5$	$8+2^2+3$	$8+2^2+3^2$	$8+2^2+3$	$8^2+2+9+3+7$	$8+2+3$	$8+2^2+3$
19	1	$4+2^2+3$	$4+2+3+11$	$4+2^5+3+11$	$8+2+3+11$	$32+8+3+11$	$8+2+3+11$	$8+2+3+11$	$8+2+3+11$	$8+2+3^2+11$

Table B1. Toda table, sphere dimensions 1-10.

k	n=11	n=12	n=13	n=14	n=15	n=16	n=17	n=18	n=19	n=20	stable
0	infy	infy	infy	infy	infy	infy	infy	infy	infy	infy	infy
1	2	2	2	2	2	2	2	2	2	2	2
2	2	2	2	2	2	2	2	2	2	2	2
3	8+3	8+3	8+3	8+3	8+3	8+3	8+3	8+3	8+3	8+3	8+3
4	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1
6	2	2	2	2	2	2	2	2	2	2	2
7	16+3+5	16+3+5	16+3+5	16+3+5	16+3+5	16+3+5	16+3+5	16+3+5	16+3+5	16+3+5	16+3+5
8	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2
9	2 [^] 3	2 [^] 3	2 [^] 3	2 [^] 3	2 [^] 3	2 [^] 3	2 [^] 3	2 [^] 3	2 [^] 3	2 [^] 3	2 [^] 3
10	2 [^] 2+3	2+3	2+3	2+3	2+3	2+3	2+3	2+3	2+3	2+3	2+3
11	8+9+7	infy+8+9+7	8+9+7	8+9+7	8+9+7	8+9+7	8+9+7	8+9+7	8+9+7	8+9+7	8+9+7
12	2	2 [^] 2	2	1	1	1	1	1	1	1	1
13	2 [^] 2+3	2 [^] 2+3	2+3	infy+3	3	3	3	3	3	3	3
14	16+2	16+4+2+3	16+2	8+2	4+2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2
15	16+2+3+5	16+2+3+5	32+2+3+5	32+2+3+5	32+2+3+5	infy+32+2+3+5	32+2+3+5	32+2+3+5	32+2+3+5	32+2+3+5	32+2+3+5
16	2	2	2	8+2+3	2 [^] 3	2 [^] 4	2 [^] 3	2 [^] 2	2 [^] 2	2 [^] 2	2 [^] 2
17	2 [^] 3	2 [^] 4	2 [^] 4	2 [^] 4	2 [^] 5	2 [^] 6	2 [^] 5	infy+2 [^] 4	2 [^] 4	2 [^] 4	2 [^] 4
18	8+4+2	32+4 [^] 2+2+3+5	8 [^] 2+2	8 [^] 2+2	8 [^] 2+2	8 [^] 3+2+3	8 [^] 2+2	8+4+2	8+2 [^] 2	8+2	8+2
19	8+2 [^] 3+3+11	8+2 [^] 5+3+11	8+2 [^] 3+3+11	8+4+2+3+11	8+2 [^] 2+3+11	8+2 [^] 2+3+11	8+2 [^] 2+3+11	8+2+3+11	8+2+3+11	infy+8+2+3+11	8+2+3+11

Table B2. Toda table, sphere dimensions 11-20 and the stable value.

Appendix C. The unstable 33-stem at the prime 2

This appendix isolates the current frontier and source inconsistency. Values are localized at 2. The $n=34$ entry contains the 2-local integers $Z_{-}(2)$, reflecting the free summand in the relevant near-stable degree.

n	$\pi_{\{n+33\}(S^n)_{-}(2)}$	Source/status
2	$Z/4 + (Z/2)^2$	Oda; summarized by Yang-Wu
3	$(Z/2)^3$	Oda; summarized by Yang-Wu
4	$(Z/8)^2 + (Z/2)^6$	Oda; summarized by Yang-Wu
5	$(Z/2)^4$	Oda; summarized by Yang-Wu
6	$Z/4 + (Z/2)^6$	Yang-Wu 2026
7	$Z/4 + (Z/2)^4$	Yang-Wu 2026
8	$Z/8 + Z/4 + (Z/2)^6$	Yang-Wu 2026
9	$(Z/2)^6$	Miyauchi-Mukai 2017
27	unresolved / source discrepancy	Yang-Wu text says project covers 10-27; caption says $n \geq 27$
28	$(Z/2)^5$	Thomeier backward theorem + stable 33-stem
29	$(Z/2)^5$	same
30	$(Z/2)^5$	same
31	$(Z/2)^6$	same
32	$(Z/2)^7$	same
33	$(Z/2)^6$	same
34	$(Z/2)^5 + Z_{-}(2)$	same
35	$(Z/2)^5$	stable and all $n \geq 35$

Source discrepancy at $n=27$

Yang-Wu's prose says the project in progress covers $10 \leq n \leq 27$; its displayed table begins at $n=28$; its caption and corollary say $n \geq 27$. This report treats $n=27$ as unresolved pending clarification.

Appendix D. Named classes and recurring notation

Notation	Stem/degree	Meaning
η	1	Hopf map; order 2; height-one.
$\eta^2=\theta_1$	2	First Kervaire class.
ν	3	Stable Hopf class; integral order 24, 2-primary order 8.
$\nu^2=\theta_2$	6	Second Kervaire class.
σ	7	Stable Hopf class; integral order 240, 2-primary order 16.
$\sigma^2=\theta_3$	14	Third Kervaire class.
$\epsilon, \mu, \kappa, \kappa\text{-bar}$	Various low stems	Toda/Mahowald named 2-primary classes; conventions vary by source.
θ_j	$2^{j+1}-2$	Kervaire-invariant-one class when it exists.
α_r	$2(p-1)r-1$, variants	Height-one/Greek-letter family at odd p .
$\beta\text{-family}$	Height-two degrees	Greek-letter height-two classes; indexing and divided variants vary.
ν_n	Period $2(p^n-1)$, up to powers and conventions	Chromatic periodicity generator in coefficient theories.
$J\text{-image}$	Bott-periodic degrees	Framed spheres from stable vector bundles.
h_i	Adams Ext class	Potential Hopf class on the 1-line; h_i^2 is Kervaire candidate.
τ, λ	Motivic/synthetic deformation parameters	Track passage from algebraic associated graded data to classical homotopy.

Named-element notation is not globally canonical. Bars, primes, subscripts, and suspensions may encode chosen unstable lifts. Before composing classes from different sources, verify the author's conventions and indeterminacy statements.

Appendix E. Source-by-source computation ledger

Source	Range/object	What it contains	Caveat/status
Toda 1962	Integral unstable/stable $k \leq 19$	Groups, generators, compositions, Toda brackets	Foundational; standard notation.
Mimura-Toda 1963	Integral $k=20$	All sphere dimensions	Completes integral all-sphere frontier.
Oda 1979	2-primary $k=25-31$; partial 32-33	Generator-level EHP and extensions	Some extensions corrected/resolved later.
Miyauchi-Mukai 2017	2-primary $k=32$; one $k=33$ group	All n for $k=32$	Current completion of 32-stem.
Yang-Wu 2026	Three $k=33$ groups	$n=6,7,8$; compiled table	Internal $n=27$ range inconsistency.
Ravenel 1986	Stable p -local, especially $p=3,5$	ANSS/chromatic tables and families	Core odd-prime reference.
Isaksen 2019	2-complete stable through 59	Adams/ANSS charts and extensions	Motivic method.
Wang-Xu 2017	Stable stem 61	Vanishing and Kahn-Priddy differential	Peer-reviewed.
IWX 2023	Stable 0-90	Exact through most range; explicit alternatives	Base of Appendix A.
Burklund-Isaksen-Xu 2025	Stable 70,71,82,83	Corrections and completions	Peer-reviewed update.
Lin-Wang-Xu 2025	Stem 126 element	Kervaire θ_6 and machine companion	Preprint.
BHLS 2023	All heights $n \geq 2$	Telescope counterexamples	Preprint.
BBQ / Carrick-Davies 2024-26	Infinite height-two families	Localized detection, torsion, products	Published/preprint mix.
Bruner-Rognes / IWX Zenodo / LWX Zenodo	Machine data	Ext, charts, finite spectra, maps	Version-specific reproducibility assets.

Appendix F. Data files accompanying this report

The report is accompanied by three machine-readable files:

- `stable_stems_0_90.csv` - corrected stable additive groups, component columns, and uncertainty status.
- `toda_unstable_stems_0_19.csv` - Toda's compact integral unstable table.
- `homotopy_spheres_bibliography.bib` - BibTeX-like bibliography for the principal sources in this review.

The CSVs are intended as research aids, not substitutes for the primary sources. In particular, they do not encode generators, products, filtrations, or Toda-bracket indeterminacies.